

AN EXPLORATION OF KNOWLEDGE DIFFUSION
THROUGH ORGANIZATIONAL BOUNDARY-SPANNING
AND EMPLOYEE MOBILITY

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A dissertation submitted to the
Faculty of the Leeds School of Business
University of Colorado in partial fulfillment
of the requirement for the degree of
Doctor of Philosophy
Strategy, Entrepreneurship and Operations Division

2019

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ABSTRACT

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An Exploration of Knowledge Diffusion through Organizational Boundary-Spanning and Employee Mobility

Dissertation directed by Professor Jeffrey J. Reuer,

In this dissertation, I attempt to study the antecedents, processes, and implications of knowledge acquisition and diffusion in the context of organizational boundary-spanning and employee mobility. In my first essay, I propose that the information intermediated by the acquisition target privately and directly reduces information asymmetry between the acquiring firm and its inherited alliance network. I demonstrate that the acquiring firm is more likely to direct its knowledge search towards the inherited alliance partners of the acquired firm as the acquirer can gain more information about them through the acquisition. In my second essay, I further propose that in an alliance between a client firm and an R&D firm, the client firm's internal structure of R&D plays an important role in the knowledge diffusion process of the alliance, specifically in determining the technological sourcing strategy in relation to their alliance partners. In my third essay, I explore the inter-firm ties as signals and their effect on inventor mobility. Based on signaling theory, inter-organizational relations such as venture capital prominence and alliance network prominence often serve as signals that boost the

visibility of the firm in strategic factor markets. This study explores the knowledge worker mobility related to signals and points to a potential downside of signaling. Overall, my dissertation essays create a link between two important strategy topics on strategic alliances and employee mobility and contribute to the research discussions in both fields.

Keywords: employee mobility, boundary spanning, knowledge search, strategic alliances, organizational structure, R&D centralization, innovation, patents

ACKNOWLEDGEMENTS

This dissertation is dedicated to my family, friends, and all those who support me in my life. First and foremost, I am grateful to my wife and my parents for their patience, understand, and continuous support throughout my years spent away and afar from them in the United States.

During such time, I have benefited greatly from the support of many people. Above all, I would like to thank my lead advisor Jeff Reuer as well as all my committee members: Janet Bercovitz, Ashton Hawk, Jin-Hyuk Kim, and Tony Tong. Their command of the literature, their wisdom, and their feedback have been the bedrock of my education. They have guided me through complex terrain with patience, and I owe them my enduring gratitude.

I would also like to thank my colleagues and friends at University of Colorado for their feedback and support at critical stages in the development of this dissertation: Stefanie Ungphakorn Cowan, Harsha Dutt Dev Tadikonda, Francisco Morales, Sandip Bisui, Aneta Oleksiak, Yuchen Zhang, Yang Yang, Aung and Poon. Without them, I would not have been able to finish my work and enjoy my experience at CU Boulder.

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CHAPTER 1.
INTRODUCTION

How does knowledge exchange, transfer, and diffuse across and within organizations? In this dissertation, I develop new theory and evidence to argue that organizational boundary-spanning through M&As and strategic alliances is a central consideration in how firms create, and then capture, valuable knowledge from human assets. I examine the strategic implications of knowledge diffusion and spillover. Specifically, I investigate how firms and investors anticipate the flow of tacit knowledge in strategic contexts such as mergers and acquisitions (M&As), strategic alliances, research and development (R&D), and the venture capital investment. Across three studies, I find evidence that knowledge spillover occurs through organizational boundary-spanning in discretely and unexpected ways. Theory and evidence also suggest that researchers and practitioners should be cautious about the strategic implications of inter- and intra-organizational shifts as well as human capital mobility. These findings have important implications for information economics and signaling, knowledge-based view of the firm, as well as for strategy-based recommendations regarding the management of human and knowledge assets.

In this introduction, I position my research agenda into a broader view of knowledge exploration through extended inter-organizational shifts, knowledge diffusion based on internal organizational structure of R&D, and knowledge spillover through employee mobility. I develop evidence that both inter- and intra-organizational strategic choices have long-term consequences for knowledge appropriation. Overall, the objective of this dissertation is to demonstrate how

employee mobility, organizational boundary spanning, and the organizational structure of R&D, affect the appropriation of value from knowledge.

Beyond Local Search: Exploring Post-Acquisition Knowledge Search between Acquirer and Inherited Alliance Partners

Extant literature on acquisitions mostly focuses on accessing the acquired firm's resources and capabilities, while ignoring the resources and capabilities from the extended network of the acquired firm. However, knowledge from this extended network is important as it enables the firm to search beyond its organizational boundaries and knowledge scope. Few studies have analyzed the acquiring firm's post-acquisition search behavior related to the acquired firm's alliance partners. In this study, we suggest that the information intermediated by the acquisition target privately and directly reduces information asymmetry between the acquiring firm and its inherited alliance network. We demonstrate that the acquiring firm is more likely to direct its knowledge search towards the inherited alliance partners of the acquired firm as the acquirer can gain more information about them through the acquisition.

Organizational Structure Matters: An Analysis of R&D Centralization and Knowledge Search in Strategic Alliances

We bridge current streams of corporate and cooperative strategy research to explore the interplay between the organizational structure of R&D and external knowledge search. We propose that in an alliance between a client firm and an R&D firm, the client firm's internal structure of R&D plays an important role in the

knowledge diffusion process of the alliance, specifically in determining the technological sourcing strategy in relation to their alliance partners. Using non-equity alliances data from the bio-pharmaceutical industry, we find that firms with centralized R&D structure tend to search more extensively from alliance partners. We further find that centralized firms conduct knowledge search more broadly from their alliance partners, and they are more likely to use such externally sourced technologies in a wider range of therapeutic areas.

A Signaling Perspective on Employee Mobility: The Darkside of Signaling

Strategy and entrepreneurship scholars have identified many benefits for new ventures to employ signals to access resources in financial and partner selection factor markets. However, scholars have not explored the signaling effect on the factor market for human capital. This paper investigates inventor mobility across biopharmaceutical ventures and examines the effects of two signal variables, VC prominence and alliance network prominence. We suggest that VC prominence and alliance network prominence can provide a higher level of visibility in the presence of information asymmetry between the new venture employees and other players in the labor market, thereby facilitating inventors changing employment owing to enhanced labor market efficiency. Empirical evidence show that biopharmaceutical ventures associated with interfirm ties such as VC and alliance signals may risk losing valuable existing human capital through employee mobility. Our study points to a downside of signaling as the employees who leave the technology venture would carry and potentially leak important tacit knowledge to their subsequent employers.

In addition, we find that the signaling effects attenuate as information asymmetry surrounding the new venture diminishes.

Evidence from Three Essays

In developing our arguments in the first essay, we build on the insight that ventures face issues of uncertainty and asymmetric information (Reuer et al., 2002; Reuer & Devarakonda, 2017), and interorganizational relationships help the acquirer shape its path of external knowledge sourcing. We argue that the acquired firm may serve as an information intermediary between the acquirer and the target firm's alliance partners, thus resolving the challenge of information asymmetry in the presence of high technological uncertainty. Therefore, the acquirer has a greater ease of accessing the knowledge of the target's alliance partners and an opportunity to broaden its search efforts beyond its organizational and technological boundaries, closer towards the search domain of the target's pre-acquisition alliance network. Probing the logic underlying our argument, we examine several contingencies - technological relatedness, prior and indirect ties between the acquirer and the alliance partners - that further enable us to unpack the information intermediation mechanism. We test our hypotheses using data on M&As and strategic alliances in the biotechnology fields. Specifically, we create a sample of firms that have engaged in acquisitions, and for each acquisition, we obtain alliances information of the acquired firm. In this way, for each acquiring firm with its associated acquisition target, we obtain a list of its inherited alliance partners. We then employ a negative

binomial framework and run regressions on the acquiring firm's backward citations towards its inherited alliance partners.

In the second essay, we develop and test our hypotheses on a dyadic sample of 960 non-equity R&D bio-pharmaceutical alliances. We explore the relationship between centralized R&D structure of the pharmaceutical firms and their external knowledge sourcing strategies their alliance partners. Our empirical context is the bio-pharmaceutical industry, in which pharmaceutical firms (referred to as the client firms) often form alliances with relatively small biotech firms (referred to as R&D firms). Due to the fact that around 86% (source: ReCap) alliances signed in the bio-pharmaceutical industry are not equity-based, we focus our study on non-equity alliances from the client firm's perspective, and examine how the internal structure of the client firms exert impact on the client firm's knowledge search strategy related to its alliance partners. From our analyses of the 960 non-equity alliances for the time period 1990-2011, we find that firms that have centralized R&D structure conduct knowledge search more heavily from their alliance partners, that they tend to conduct broader knowledge search by citing a wider range of technological classes from their partners, and that they use such cited knowledge in a wider range of therapeutic areas.

In the third essay, we conduct empirical analyses in the context of inventor movements in the biopharmaceutical industry. This setting offers several advantages: First, inventors are specialized human capital which is of great importance to the firm. Second, while change of employment is common in the

industry, the use of patent data enable use to track inventor movements relatively accurately. Third, there are high frequencies of VC investments and alliance formations which enable us to observe the signaling events using network measures. Our empirical findings on negative binomial regressions largely support our hypotheses and demonstrate that both signals of VC prominence and alliance network prominence are associated with existing inventors subsequently leaving the focal venture. In addition, our data reveals that these signals are more pronounced when there is a high level of information asymmetry as we find that the signaling effects are reduced with the number of investment rounds.

Contributions

In the first essay, our study reveals a thus-far understudied area in M&A literature. By investigating the roles of the target's pre-acquisition alliance network bestowed upon the acquirer, we extend existing M&A research to offer a unique perspective of information intermediation. We suggest that the acquired firm serves to reduce information asymmetry between the acquirer and distant technology ventures, and it directs the acquirer's search efforts towards the search domain of the target's alliance network. In this regard, our paper extends previous research about how acquisitions provide the acquirers with access to an extended network of knowledge beyond the scope of their local knowledge search (Rosenkopf & Nerkar, 2001; Zollo & Singh, 2004; Puranam et al., 2006). We also extend alliance literature and explore the benefits of the inherited alliance network. Existing alliance research has mostly overlooked this knowledge benefit of alliance networks towards the acquisition partner (Zollo and Singh, 2004; Puranam et al., 2006). We point to

another mode of configurational partner change where the acquiring firm inherits the alliances of the acquired firm. We suggest that the extended alliance network of the acquired firm provides the acquirers with additional benefits beyond the acquired firm boundary. We demonstrate that the acquirer becomes aware of the acquired firm's alliance network after going through an acquisition, and their subsequent search direction is pulled toward these inherited alliances.

In the second essay, we contribute to the organizational structure research and bridge it with research on organizational boundary-spanning, strategic alliances, and knowledge search. Our paper complements existing studies aimed at analyzing the link between R&D centralization and external knowledge acquisition. Firms typically achieve external knowledge acquisition through M&As and strategic alliances (Rosenkopf and Almeida, 2003), and it is often assumed that firms would easily combine such acquired knowledge with its existing knowledge stock. However, as we consider the organization's internal knowledge repository and that knowledge is often segregated across different research units and teams ((Daneels, 2011)), this study extends our understanding of knowledge acquisition and recombination within the organizational structure. We also offer insights on how centralization of R&D within a firm can exert impact on the client firm's knowledge search behavior towards its strategic alliance partners. By considering how pharmaceutical firms manage R&D and capabilities internally across research units, we find that R&D centralization plays an important role in shaping the client firm's external knowledge search trajectory towards to its alliance partners (Arora and Gambardella, 1994;

Zheng et al., 2007). Drawing upon both alliance and patent data, our results show that firms with centralized R&D are focused on higher level innovation and thus are more likely to diffuse and recombine external alliance partner knowledge more effectively within their own research units. Besides, as empirical studies on organizational structures are rare, we provide empirical evidence using data in biopharmaceutical industry to proxy for the R&D structure of the firm. We measure R&D centralization based on whether the position of an R&D head is present in the top management team (Lerner and Wulf, 2007). Our findings are also in line with prior research on R&D centralization using alternative measures (Argyres and Silverman, 2004; Arora et al., 2014), demonstrating that firms with centralized R&D are associated with better communication efficiency when it comes to R&D coordination and are more likely to generate innovations with a wider breadth of technological classes and therapeutic areas.

In the third essay, our study makes contributions to the literature on signaling and strategic human capital retention for new ventures. This paper first contributes to the entrepreneurship literature by showing how new ventures may face challenges in keeping important human capital with signaling, and we contribute empirical evidence by showing that new ventures' network affiliations negatively affect the likelihood that new employees are going to stay in the newly established firms. Second, we build an underexplored link between firm level signaling and human capital retention. While Spence (1973) investigates job candidates signaling to hiring firms, our study examines the counterpart of the employment relationship where new

ventures engage in signaling through interorganizational ties to attract potential employees in the labor market. Second, while prior studies on signaling theory have identified many benefits of signals, our study points to a potential downside of signaling which has been largely overlooked. Third, signaling theory research has also suggested that the benefits of interorganizational relationships are likely to depend on the informational environment of signals in different stages of the life of the firm (e.g. Gulati and Higgins, 2003). Our study examines the signaling effects over time and finds that these effects become less pronounced as the new venture goes through multiple investment rounds.

CHAPTER 2.

Beyond Local Search: Exploring Post-Acquisition Knowledge Search between Acquirer and Inherited Alliance Partners

Introduction

In high-tech industries, firm success often depends on its innovation process and technology trajectory (Nonaka & Takeuchi, 1995; Leonard-Barton, 1995). Yet, high-tech ventures often face constraints while searching for knowledge due to their organizational and technological boundaries (Rosenkopf & Nerkar, 2001; Nelson and Winter, 1982; March and Simon, 1958). Existing studies have shown that, in order to go beyond local search, firms must innovate in part through external knowledge acquisition, in the form of acquisitions (Ahuja and Katila, 2001; Puranam et al., 2006) and strategic alliances (Grant & Baden-Fuller, 2004; Mowery et al., 1996). However, research on knowledge search has stopped short of investigating how the combination of acquisitions and strategic alliances might affect high-tech ventures' knowledge search and innovation activities.

To access external knowledge sources, firms search beyond their organizational boundaries through both M&As and strategic alliances (Cohen & Levinthal, 1990; Song, Almeida, & Wu, 2003). The ability to produce multiple product innovations in quick succession is critical in fast-changing environments (Brown & Eisenhardt, 1997). Many companies adopt external knowledge acquisition to avoid the time-consuming, path-dependent, and uncertain processes of internally accumulating capabilities for producing streams of innovation (Dierickx & Cool, 1989; Leonard-Barton, 1995). On the one hand, M&As have become prevalent vehicles for obtaining resources in the knowledge intensive industry existing studies typically assumed that the acquired firm knowledge is embedded within its firm boundary (Chaudhuri & Tabrizi, 1999). As an increasing number of firms engage in M&As to

gain access to the knowledge base of the acquired targets, many researchers have scrutinized how knowledge of an acquired firm affects an acquirer's innovation performance (Ahuja & Katila, 2001). In an acquisition, while the acquiring firm must integrate the acquired firm to some extent in order to absorb and combine its technology with its existing knowledge stock, at the same time, the acquirer would face difficulty to preserve the acquired firm's capacity for continued innovation (Puranam, Singh, & Zollo, 2006). Though the acquiring firm attempts to capture all resources and innovation capabilities within the acquired firm, the acquisition tends to disrupt the acquired firm's innovation trajectory due to changes in its existing organizational routines and its extended collaboration network.

On the other hand, firms may also choose to collaborate and access external resources through strategic alliances (Mowery, Oxley & Silverman, 1996; Grant & Baden-Fuller, 2004). Empirical studies of alliances have shown that they are useful mechanisms for knowledge acquisition and learning, and the collaboration between alliance partners is essential for innovation (Hamel et al. 1989; Doz, 1996; Dyer, 1997; Reuer & Arino, 2007). In particular, scholars have identified that the firm's alliance network plays an important role in a firm's innovation process (Dyer & Singh, 1998; Gulati & Gargiulo, 1999). They help firms combine their know-how and expertise with various resources of their alliance partners to generate new technologies and augment their innovative activities (Powell et al., 1996; Rothaermel & Deeds, 2004). However, the resources involved in R&D collaboration are mostly intangible, and firms may have to perform costly search and evaluation to identify prospective

partner's technological assets and development capabilities (Stigler, 1961; Rangan, 2000; Vanhaverbeke et al., 2002; Villalonga & McGahan, 2005). In addition, prospective partners may distort information about their R&D resources (Arrow, 1962; Teece, 1986; Dushnitsky & Shaver, 2009), such as by misrepresenting the quality of their technological resources and prospects (Gulati & Gargiulo, 1999; Nicholson, Danzon & McCullough). This in turn would further raise the cost of knowledge search of the non-partnering firms, unless these firms are able to overcome the associated information asymmetries.

Although existing research demonstrate the importance of M&As and alliances with knowledge acquisition, it has stopped short of addressing them synthetically and investigating how organizational boundary-spanning may affect subsequent technological boundary-spanning. This study draws on a comprehensive perspective and examines empirically the relationship between M&As, strategic alliances, and the acquirer's subsequent knowledge search trajectory. Given the combined settings of M&As and R&D alliances, we suggest that technology-based ventures would search beyond the scope of the acquired firm and dive into the knowledge network of the target's alliance partners as well. This study elucidates the interplay of organizational and technological boundary-spanning in the context of acquisitions and the inherited alliance network of the acquired firm. First, for a technology-based venture to attain external knowledge, acquisitions span organizational boundary and overcomes the trap of local search. Further, the acquisition target facilitates

technological boundary-spanning and directs the tech venture towards the knowledge network of the acquired firm's alliance partners.

In developing our arguments, we build on the insight that ventures face issues of uncertainty and asymmetric information (Reuer et al., 2002; Reuer & Devarakonda, 2017), and interorganizational relationships help the acquirer shape its path of external knowledge sourcing. We argue that the acquired firm may serve as an information intermediary between the acquirer and the target firm's alliance partners, thus resolving the challenge of information asymmetry in the presence of high technological uncertainty. Therefore, the acquirer has a greater ease of accessing the knowledge of the target's alliance partners and an opportunity to broaden its search efforts beyond its organizational and technological boundaries, closer towards the search domain of the target's pre-acquisition alliance network. Probing the logic underlying our argument, we examine several contingencies - technological relatedness, prior and indirect ties between the acquirer and the alliance partners - that further enable us to unpack the information intermediation mechanism.

We test our hypotheses using data on M&As and strategic alliances in the biotechnology fields. Several features of this setting are relevant to our study. First, biotech ventures explore discoveries in genomics and use modern genetic techniques to identify new targets for drug therapies, in ways that depart quite fundamentally from the chemical methods that pharmaceutical firms have traditionally relied upon to discover new drugs (Pisano 1990, Shan et al. 1994, Drews 2003). As a result,

biotech-biotech deals are mostly focused on external knowledge acquisition and search, rather than obtaining access to downstream resources and activities, which is more suitable for our study setting. Further, as biotech ventures are constantly innovating, collaborations through R&D alliances are prevalent and often combined with acquisitions, which makes it a perfect setting for studying the intersection of organizational and technological boundary spanning. Finally, the USPTO has a sufficient amount of patent records in the biotechnology fields that enable us to investigate the search behaviors of the biotech ventures.

This study reveals a thus-far understudied area in M&A literature. First, by investigating the roles of the target's pre-acquisition alliance network bestowed upon the acquirer, we extend existing M&A research to offer a unique perspective of information intermediation. We suggest that the acquired firm serves to reduce information asymmetry between the acquirer and distant technology ventures, and it directs the acquirer's search efforts towards the search domain of the target's alliance network. In this regard, our paper extends previous research about how acquisitions provide the acquirers with access to an extended network of knowledge beyond the scope of their local knowledge search (Rosenkopf & Nerkar, 2001; Zollo & Singh, 2004; Puranam et al., 2006).

Second, we extend alliance literature and explore the benefits of the inherited alliance network. Existing alliance research has mostly overlooked this knowledge benefit of alliance networks towards the acquisition partner (Zollo and Singh, 2004; Puranam et al., 2006). We point to another mode of configurational partner change

where the acquiring firm inherits the alliances of the acquired firm. We suggest that the extended alliance network of the acquired firm provides the acquirers with additional benefits beyond the acquired firm boundary. We demonstrate that the acquirer becomes aware of the acquired firm's alliance network after going through an acquisition, and their subsequent search direction is pulled toward these inherited alliances.

Finally, we complement research that has focused on other ties or intermediaries between a venture and its potential knowledge search targets in the market for technology (e.g. Raub & Weesie, 1990; Gulati, 1995; Mowery et al., 1996; Lane & Lubatkin, 1998; Stuart, 1998; Gulati & Gargiulo, 1999). We build upon and extend this stream of research by demonstrating that the effects of information intermediation through acquisitions are greatest when acquiring firms have relatively low level of technological relatedness or lack familiarity with the extend network of the acquired firms' alliance partners.

Theory and Hypotheses

Firms rely on acquisitions to gain valuable resources that are hard to acquire via market exchanges (Capron, Dussage, & Mitchell, 1998; Capron, 1999). Organizational knowledge built internally through experience is a notable example of such resources (Kogut & Zander, 1992; Song, Almeida, & Wu, 2003). In this respect, M&As came to force as an effective strategy to acquire new knowledge which is difficult to obtain through market transactions. Accordingly, many researchers have examined how knowledge within the acquired firm improves the acquirer's

innovation performance. Ahuja and Katila (2001), for example, suggested that by absorbing the new knowledge base of an acquired firm, an acquirer can increase the potential for an inventive recombination, thereby enhancing its innovation outputs through technological acquisitions. Prior studies of knowledge-seeking M&As also examined knowledge relatedness and complementarity between the acquiring firms and acquired firms, with a focus on how an acquisition enables the acquirer to assimilate the knowledge within the acquired firm (e.g. Makri, Hitt, & Lane, 2010). Evidently, M&As can be an effective means for the acquirer to absorb knowledge that resides in the acquired firm.

A firm's critical resources, however, may span beyond firm boundaries (Dyer & Singh, 1998). Since a firm's knowledge stock is accumulated through interactions among diverse external organizations, a breakthrough innovation is hardly achieved solely based on the knowledge within a single firm (Badaracco, 1991). Organizations depend on the inter-organizational cooperation to access resources that are essential to achieve their goals but under the control of other organizations (Gulati & Gargiulo, 1999; Pfeffer & Salancik, 1978). Recent empirical studies have shown the contribution of R&D alliances to the firm's knowledge base, knowledge recombination, and innovation performance (Baum, Calabrese, & Silverman, 2000; Ahuja, 2000; Rothaermel & Hess, 2007). Scholars state that innovation process should be understood as a comprehensive mechanism which includes the focal firm's relationship with other organizations through which the firm can access external resources. In this paper, we focus on an acquired firm's knowledge which may reside

in its alliance relationships in order to capture the real value of the acquired firm's resources.

Considering that alliance partners are valuable resources, the acquiring firm's post-acquisition performance depends on not only knowledge within the acquired firm but also knowledge across its alliance network. That is, an acquirer can benefit from the target firm's alliance network. Anand, Capron, & Mitchell (2005) showed that extra benefits arise from gaining access to the networks of the acquired firms with multinational networks in their study of cross-border acquisitions. This is because the network of the acquired firm provides the acquirer with diverse resources that reside in its established multinational network. In a similar vein, we suggest that the alliance partner network of an acquired firm can also be beneficial to an acquiring firm's knowledge base and innovation performance, if the acquirer facilitates interactions and knowledge diffusions among the firms engaged in both firms' networks.

After acquisitions, acquirers become more aware of the acquired firm's extended network of alliance partners. Acquired firms serve as an information intermediary between the acquirer and the inherited alliance partners. Since not only direct but also indirect linkages play an important role in the innovation process by collecting, processing, and transmitting knowledge between organizations (Ahuja, 2000; Salman & Saives, 2005), the close linkages generated through the acquisition will facilitate knowledge exchange and recombination among the partners. Gulati (1995) also showed that a shorter path length between two firms in a global network

increases the possibility of a new alliance formation with each other. Therefore, the alliance network of an acquired firm will have an impact on the acquiring firm's innovation process, interacting with the acquiring firm and its partners.

We expect that, in the context of inherited alliances, the acquired firm can have an important mediating role in navigating the relationship between the acquiring firm and the inherited alliance partner. Specifically, it has the potential to act as an information broker that transfers knowledge and provides information linkages between the acquiring firm and the inherited alliance partner. Although the acquiring firm becomes the formal partner in the alliance after the acquisition, and it inherits the extended network of the target alliance partners, which gives it the possibility to broker relevant knowledge between the acquirer and the inherited alliance partners. The acquirer's understanding of the technological resources and capabilities of its inherited alliance partners allows it to search for information and technical know-how about the inherited alliance to the acquiring firm (Aldrich & Herker, 1977). Over the course of the alliance, the acquired firm has also developed knowledge about how the alliance operates (Schreiner, Kale, & Corsten, 2009). The acquired firm can therefore transfer information to the acquiring firm about what processes are implemented for realizing particular alliance objectives. In sum, we suggest that the acquirers can search for relevant information on the inherited alliance, consolidate and recombine its knowledge in an efficient way from the inherited alliance partners with relatively low levels of information and search costs.

H1. After an acquisition, the acquirer is more likely to direct its knowledge search towards the alliance partners of the acquired firm.

While the above argument indicates that the acquired alliance partners can play an important role in addressing the core challenges of managing inherited alliances, we expect that the ability of the target firm to execute the information intermediary role depends on certain conditions between the acquiring firm and the inherited alliance partner, i.e. their technological relatedness, history of joint collaboration, as well as their indirect association. We develop hypotheses on how these conditions influence the search behaviors of the acquiring firm in the paragraphs below.

Technological relatedness refers to the extent to which the knowledge bases of alliance partners cover similar domains, and the existence of common technological resources between prospective R&D alliance partners can directly reduce information asymmetry between them. Recognizing the potential for the achievement of collaboration goals by partners with a common pool of technological endowments is easier than for those drawing from entirely distant knowledge bases (Lane & Lubatkin, 1998). Firms that possess common technological resources are also better able to understand and evaluate the capabilities of a potential R&D partner because of their mutual understanding of shared technological bases (Schildt & Laamanen, 2006). For example, one way in which firms in high-tech industries develop common technological bases over time is by drawing from the common pool of patents of other firms in the industry (Mowery et al., 1996; Lane & Lubatkin, 1998). Through past

citations to the same third-party patents, prospective partners can reduce costs of due diligence to evaluate each other's critical resources (Stuart, 1998) because they already possess knowledge about each other's patents and technological endowments relevant to potential collaborations. However, when such knowledge about potential R&D partners is lacking, information mediated by the acquired firm in the acquisition becomes particularly valuable.

Firms operating in highly similar technological domains will be able to evaluate which resources are needed. Prospective R&D alliance partners that are familiar with each other's technological resources in particular have the benefit of reduced information asymmetry between them. When firms have developed such a direct understanding about potential R&D partners' technological endowments, they are better equipped to evaluate the underlying quality of their resources and mitigate the risk of adverse selection in a prospective partnership (e.g., Cohen & Levinthal, 1990; Rothaermel & Boeker, 2008). For example, one way in which potential partners become familiar with each other's technologies is by mutually drawing from each other's knowledge resources through activities such as patenting (e.g., Mowery et al., 1998). More specifically, by virtue of patent citations prospective partners may benefit in several ways. Over time they accumulate insights into a potential partner's technology development trajectory which can help them assess prospects for future collaborations (e.g., Cohen & Levinthal, 1990; Gilsing et al., 2008). They can also reduce costs of due diligence to evaluate each other's critical resources and knowledge stock (e.g., Stuart, 1998), which can be significant in high-tech industries. When firms

become familiar with partners' technological resources and capabilities through their patenting activities, they can also make more accurate assessments of the difficult-to-evaluate intangible R&D resources that may be hard or even impossible to perform by other means (e.g., Akerlof, 1970; Nicholson et al., 2005).

The foregoing discussion suggests that technology relatedness and the acquisition target are distinct yet partially redundant sources of information about potential inherited alliance partners. Thus, we argue that the information intermediary role of the acquisition target firm become less useful in directing the acquiring firm's subsequent search behavior when the acquiring firm can directly learn about each other due to sharing common technological activities and resources. By contrast, when firms lack such technological relatedness, the effect of acquiring acquired firm on knowledge search becomes more pronounced and will have a greater bearing on inhering the acquired firm's alliance network. Taken together, we therefore posit:

H2. The technological relatedness between the acquiring firm and the alliance partners of the acquired firm negatively moderates their post-acquisition knowledge flow.

The history of prior collaboration becomes another important source of information intermediary about the reliability and capabilities of their current and potential search targets. Such information helps firms to learn about new technological opportunities and it also enhances their ability to recombine their

technical know-how. By providing conduits for valuable information, this firsthand information is particularly effective in several ways. First, firms with relational connections are likely to have greater understanding of each other's needs and capabilities and are thus likely to be one of the first to spot new opportunities for knowledge creation and recombination. Second, information about a partner based on prior interactions can reduce the hazards associated with future transactions and thus also increase the parties' interest in future ties (Zucker, 1986; Kogut, 1989; Heide & Miner, 1992; Gulati, 1995). New ties between prior partners can then result from repetitive momentum such that, once two firms enter an alliance, they do so repeatedly and therefore cite each other's patents in the future. Such repetitive behavior by organizations results from the establishment of organizational routines, and they might develop specific routines for managing an interface with each other, making it easier for them to search for innovation opportunities in the future (Cyert & March, 1963; Nelson & Winter, 1982).

Therefore, prior alliance experiences provide a firm with valuable information that it can use to seek potentially beneficial knowledge from its prior partners. In searching for subsequent knowledge in the market for technology, a natural solution for a firm is to first consider its previous partners with which it has had direct prior alliance experiences. The tendency to conduct a limited search among existing alliance network will be greater when a firm is more uncertain about the benefits it derives from the selected option (Cyert & March, 1963; Granovetter, 1985). Thus, when trustworthy partners are already available, searching for outside knowledge is

hard to be rationalized in the market for technology. Accordingly, we propose that prior alliance ties between the acquiring firm and its inherited alliance partners would substitute and negatively moderates the information intermediary role of the acquisition target. As prior alliance partners would have already developed knowledge connections, the effect of the acquisition target on the acquirer learning about knowledge from and redirecting its search towards its prior alliance partners would be significantly reduced.

H3. The history of prior ties between the acquirer and its inherited alliance partners negatively moderates the post-acquisition knowledge search of the acquiring firm towards its alliance partners.

In addition, being close to each other in the network of R&D alliances provides firms with information which enhances the likelihood of subsequent knowledge search. Above and beyond a firm's direct ties, the overall structural context in which a firm is placed in its knowledge network can also influence its choice of search. That is, indirect connections through common partners can play an important role in leading to exchange of technical know-how and information between firms. The simplest form of indirect ties is that two firms share a common partner in their separate alliance networks, and two such firms are likely to have access to more information about each other than two firms with no such connection. Like previously mentioned direct ties, indirect ties make otherwise unfamiliar firms aware of each other by serving as effective referral networks (Van de Ven, 1976; Kreps, 1990; Raub & Weesie, 1990; Portes & Sensenbrenner, 1993; Burt & Knez, 1995). Because

the indirect ties can function as an information conduit and because a common actor can play the role of a reference and become a mechanism, we expect that two firms' indirect ties through a third party may enhance the chances that the two firms will gain information and conduct knowledge search from each other's knowledge stock. Efficiency in technological interactions is more easily attained when firms receive information about their potential search targets through interaction with third parties than when they are atomized with no available information sources about each other (Raub & Weesie, 1990).

When indirect prior ties exist between an acquiring firm and its inherited alliance partners, we expect that two firms would already have prior chances to better understand and evaluate each other through mediation and information provided by the common third parties rather than the acquisition target, and thus the acquiring firm would already include them within its search domain. Therefore, we hypothesize that the information intermediating role of the acquisition target would be significantly reduced in the presence of an indirect tie between the acquiring firm and its inherited alliance partners.

H4. The indirect ties between the acquirer and its inherited alliance partners negatively moderates the post-acquisition knowledge search of the acquiring firm towards its inherited alliance partners.

Data and Method

Data sources

We test our hypotheses using acquisitions and R&D alliances data from biotechnology firms between 1992 and 2010. The biotechnology industry is appropriate for the setting of our study for several reasons. First, biotechnology firms are constantly searching for new technology, and their capability to innovate is directly linked to the venture's performance and growth (Pisano, 1990; Drews, 2003). Second, biotechnology firms are active in engaging in acquisitions and strategy alliances, which gives us a perfect setting to observe organizational boundary-spanning (Hagedoor & Duysters, 2002). Third, most innovations in the biopharmaceutical industry are recorded in patents, which makes technological boundary-spanning easily observable through information on patent assignees, inventors, as well as citations (Hall et al., 2001; Li et al., 2014).

We constructed a dataset containing acquisitions and the acquired firms' networks of alliances in the biotechnology industry. To construct our sample, we first gathered data sources of acquisition deals including *SDC Platinum Mergers & Acquisitions*, multiple corporate reporting channels (such as annual reports, SEC 10-K and 20-F filings), as well as acquisition statements (SEC 8-K filings). To obtain more complete information on the acquired firms and their alliance networks, we subsequently searched for press release information on LexisNexis database and *RECAP Biotech Alliance Database*, and we gathered active alliances information around the time of acquisition. We were also able to create dyad-level and firm-level information such as prior alliance ties, indirect ties, firm age, firm size, prior

acquisition experience, and prior alliance experiences based on these datasets. In addition, we used *BvD Orbis Database* and Google search to further obtain geographic and locational information of the aforementioned firms.

Thus, we created a sample of firms that have engaged in acquisitions, and for each acquisition, we obtained alliances information of the acquired firm. In this way, for each acquiring firm with its associated acquisition target, we obtained a list of its inherited alliance partners. Overall, we identified 107 majority acquisitions pursued by the top 30 biotechnology firms based on their 2014 revenues, and we collected information on the growth and search information of these firms over the period of 1992 and 2010. From this sample, we identified whether the networks of target alliances were inherited after acquisitions. In our final sample, we comprised 45 acquisition deals with a total of 112 identified inherited alliances.

We collected search information through patents and citation data from the *USPTO* and *Harvard Patent Network Dataverse* (Li et al., 2014). We first gathered patent information by matching patent assignee names to the firm names of acquiring firms, acquired firms, and inherited alliances respectively. We used disambiguation methods and data for patent assignees and inventor based on Li et al. (2014). To obtain knowledge search information, we matched citation information to specific patent based on patent number, and we also track inventor information pre- and post- acquisition so that we were able to distinguish patents developed by the acquiring firm inventors from those by the acquired firm inventors. To test whether the acquiring firm redirects its knowledge search towards its inherited

alliance partners after the acquisition, we track the number of backward citations coming from its corresponding inherited alliance partners by the acquiring firms. Specifically, we removed those acquiring firm patents that were invented by the original target firm inventors, so that our results wouldn't be biased due to data noise from continued research and development efforts between the acquired firm its original (inherited) alliance partners. In addition, we were able to calculate measures such as backward citations and technology relatedness using detailed patent citation data from the *USPTO*. Together with the aforementioned dyad-level and firm-level information we collected, our final sample consist of 358 observations over time.

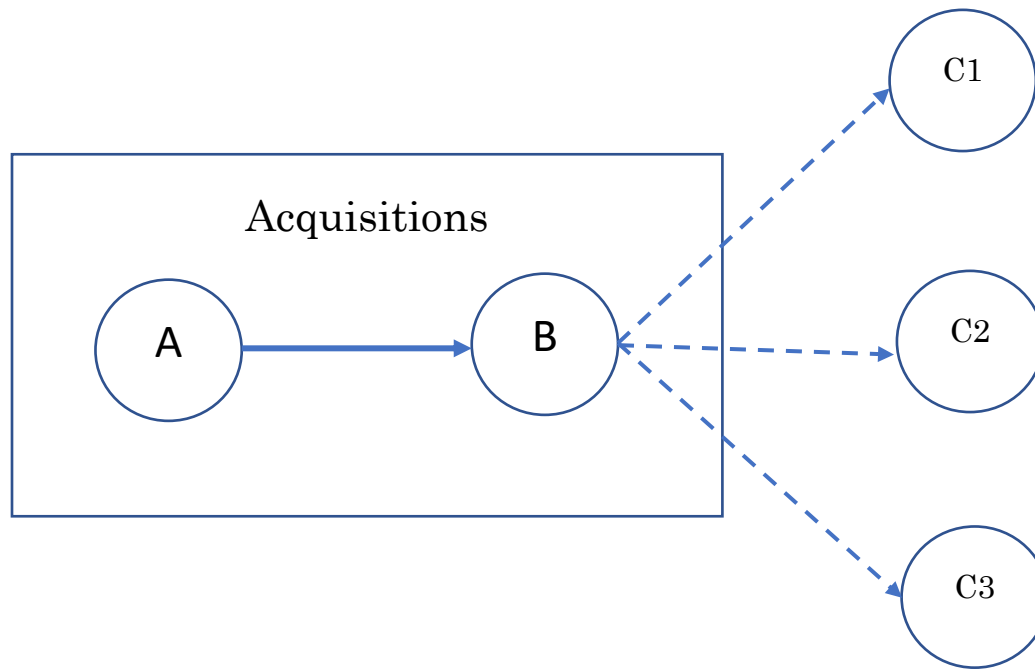
Dependent variable

Acquirer's search in the inherited alliance network. Following tradition in existing research on firms' search (e.g. Rosenkopf & Nerkar, 2001; Katila & Ahuja, 2002; Rosenkopf & Almeida, 2003), we measured a firm's search using backward citation data that the focal firm's successful patent applications make to prior patents. We first calculated the sum of citations an acquiring firm searches in its inherited alliance partner's knowledge domain in a given year. For any given year before the year of the acquisition event, we then calculated our dependent variable as the cumulative sum of citations the acquiring firm searches in the inherited alliance partner's knowledge domain up to 3 years prior to the given year. Alternative, for any given year after the year of the acquisition event, we calculated our dependent variable as the cumulative sum of citations the acquiring firm searches in its inherited alliance partner's domain up to 3 years after the given year. Thus, a greater

value in the resulting variables reflects a greater focus in the acquiring firm's search behavior towards its inherited alliance partner.

To illustrate with a generic example in Figure 1, Firm A acquires Firm B in 1995, and Firm A continues to engage in an alliance with Firm B's alliance partner, Firm C, and thus Firm C becomes an inherited alliance partner of Firm A. specifically, if an acquiring firm (Firm A) cites its inherited alliance partner (Firm C) once a year from 1992 to 1994, then the dependent variable for Firm A in 1994 would be 3, which is the cumulative sum of backward citations coming from Firm C in the previous 3 years. Alternative, if Firm A cites Firm C twice a year after the acquisition from 1996-1998, then the dependent variable for Firm A in 1996 would be 6, which is the cumulative sum of citations in the next 3 years to come. To check for robustness of the measure, we also calculated our measure using 1-year and 5-year windows instead of a 3-year window, and results are qualitatively similar.

Figure 2-1. Illustration of the Acquiring Firm and its Inherited Alliance Partners



Note: Acquiring Firm – A, Acquired Firm – B, Inherited Alliance Partner – C1, C2, C3

Independent variables

Acquisition. As Hypothesis 1 suggests, we expect that, after the acquisition, the acquiring firm is more likely to direct its knowledge search towards the alliance partners of the acquired firm. This is the key independent variable that refers to whether the acquisition has taken place for the focal dyad of acquiring firm-inherited alliance partner in the focal year. This dummy measure takes value of 0 if the focal dyad-year observation is before the year of acquisition, and it takes a value of 1 if the acquisition has already happened for the focal dyad-year. As we intend to compare the acquiring firm's post-acquisition search behavior against its pre-acquisition

search behavior, we also removed observations in the acquisition year to reduce data noise. After controlling for other potential mechanisms that are associated with the acquiring firm's knowledge search behavior, we expect that the number of backward citations coming from the inherited alliance partners would significantly increase after the year of acquisition.

Technological relatedness. Following previous research (e.g. Jaffe, 1986; Sampson, 2007), we measured the technology relatedness between the two firms in a given year as the extent to which the two firms received patent grants in the same technology classes in the 3 years prior to the given year. This allows us to capture the relative technological position of the alliance partners. We calculated this measure between *Firm i* and *Firm j* as the angular proximity of vectors F_i and F_j , where $F_i = (F_i^1, F_i^2, \dots, F_i^s)$ is the vector representing the number of patents assigned to *Firm i* in patent class s :

$$\text{Technology relatedness} = F_i F_j' / \sqrt{(F_i F_i')(F_j F_j')}$$

This measure ranges between 0 and 1, where 0 indicates no overlap and 1 indicates a complete overlap in technological base between the two firms. Given that technological relatedness reduces information asymmetry and build knowledge connections between the acquirer and the inherited alliance partner, we expect that the technological relatedness between the acquirer and the inherited alliance partners would be positively associated with the acquirer's backward citations towards the inherited alliance partners, and as Hypothesis 2 suggests, the positive

relationship between the acquisition dummy and the acquirer's backward citations towards the inherited alliance partner would be attenuated by a high level of technological relatedness between the two firms. As a robustness check, we also calculated this measure of technological relatedness using 5-year and 7-year windows instead of a 3-year window, and results are consistent.

Prior ties between acquirer and inherited alliance partner. Using the extensive alliance information from *RECAP* and *LexiNexus* databases, we tracked whether a prior alliance exists between the acquiring firm and the inherited alliance partner three years prior to the focal dyad-year (Reuer & Devarakonda, 2017). In our sample, we barely observed cases in which the acquiring firm and the inherited alliance partner formed more than one prior alliance tie during the three-year window. Thus, we coded the *prior ties* measure to be 1 if the firms have an alliance in the past three years, and 0 otherwise. Given that prior ties reduces information asymmetry and build knowledge connections between the acquirer and the inherited alliance partner, we expect that this measure of prior ties between the firms dyad would have a positively relationship with the acquirer's knowledge search towards the inherited alliance partner, and as Hypothesis 3 suggests, we expect that the positive relationship between the acquisition dummy and the acquirer's backward citations towards the inherited alliance partners would be attenuated when prior alliance ties exist between the two firms.

Indirect ties between the acquirer and inherited alliance partner. Similar to prior direct ties, we track indirect alliance relationships between the acquirer and

the inherited alliance partner in the previous three years from the focal dyad-year. Following Boyd & Spekman, 2008, we code the indirect ties to be 1 if the acquirer and the inherited alliance partner shared at least 1 common alliance partner in the previous 3 years, and 0 otherwise. Given that prior indirect ties would reduce information asymmetry and build knowledge connections between the acquirer and the inherited alliance partner, we expect that this measure of prior ties between the firms dyad would have a positively relationship with the acquirer's knowledge search towards the inherited alliance partner, and as Hypothesis 4 suggests, we expect that the positive relationship between the acquisition dummy and the acquirer's backward citations towards the inherited alliance partners would be attenuated when there existed a common alliance partner between the two firms. We exclude the acquired firm as an indirect tie in the sample, as the intention of Hypothesis 4 is to test alternative mechanisms that reduces information asymmetry between the acquirer and the inherited alliance partner apart from the focal acquisition.

Control variables

We control for a great many factors and firm characteristics related to the acquiring firm and the inherited alliance partners, as they would also have impact on the acquiring firm's knowledge search direction. First, to better understand the post-acquisition relationship between the acquiring firm and the acquisition target, we created a measure to observe whether the acquired target is preserved as a stand-alone firm after the acquisition. To obtain information regarding the target preservation, we examined press releases in the *LexisNexus* database in the first few years after the acquisition to check whether the acquired firm was continuously

traceable under its original name (Paruchuri et al., 2006; Puranam et al., 2006 & 2009). If the acquired firm frequently showed up under its original name in the press releases a year after the acquisition, then we consider it to be retained as a stand-alone operating entity. Simultaneously, we checked for annual reports and *SEC 8-K* acquisition statements to see whether the acquired firm was reported to function as a distinct operational unit after the acquisition, for instance, as a wholly owned subsidiary or a separate unit/division, to account for cases where name changes occurred for the target firm due to legal purposes, such as clinical trials (but their operations remained autonomous). In such cases, we would still consider the acquired firm as preserved. Otherwise, if the acquired firm became untraceable in neither the press releases nor company reports/statements, then we would assume that the acquired firm was structurally absorbed into the acquiring firm. Based on the above data collection efforts, we coded the *target preservation* measure as 1 if the acquired target was preserved, and 0 otherwise.

Furthermore, we control for the acquiring firm characteristics such as the *acquirer age* and the *acquirer size* (measured as the number of employees). This is because both firm age and size may have implications on how the acquiring firm manages its research and development internally. We also included the acquirer's *acquisition intensity*, which is measured as the total number of acquisitions the acquiring firm has pursued in the previous five years.

Regarding inherited alliance partners, we controlled for whether venture capital firms are investing in the focal inherited alliance partner, as *VC backing* may

have implications on both the alliance decisions as well as the firm's patenting strategies. Similarly, we also control for whether the inherited alliance partner is in fact a *large pharma* firm (=1 if yes, =0 if no) based on its revenue information in the year of the acquisition. Besides, we calculated the inherited alliance partner's alliance experience as the total number of alliances the firms has formed in the past five years, and we controlled for the firm age as well.

On the dyad level, we expect that the relative distance between the acquiring firm and its inherited alliance partners would play a role in influencing the acquiring firm's knowledge search domain, and we control for three distance measures including geographic distance, cultural distance, and industry relatedness, respectively. Additionally, as our sample has a panel structure, we include year fixed effects to control for macroeconomic conditions and other unobserved heterogeneities in the focal industry.

Analysis and Results

Due to the count nature and over dispersion of patent citations data, we employ a negative binomial framework for our regression on the acquiring firm's backward citations towards its inherited alliance partners. Table1 presents the descriptive statistics and the correlations matrix of the variables in the analysis. Our final sample consists of 358 dyad-year observations in our sample with a reasonable amount of backward citations occurring across the observation window. At a given year, the maximum number of backward citations towards the inherited alliance partner in our sample is 5, and the minimum is 0 meaning that no patent applications

have referenced from the inherited alliance partners. We also examined variance inflation factors (VIF) to diagnose multicollinearity concerns and find that the mean VIF is less than 3 for all models, which is well below the critical value of 10.

Table 2 reports results of negative binomial models where the dependent variable is the count of backward citations in the focal dyad-year observation. The dispersion parameter is significant in all our models, thus confirming over dispersion in the data and supporting the use of the negative binomial models. We report heteroskedasticity-robust standard errors clustered on the venture firms. In Table 2, column (1) is the baseline model with just control variables. Column (2) shows the main effect of the acquisition dummy without interactions, and columns (3)- (5) show interaction effects between each moderator and the acquisition dummy as alternative mechanisms for reduction of information asymmetry. Finally, columns (6) includes the full model that includes all three interaction terms.

Table 2-1. Descriptive Statistics

	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Backward citations	1.30	1.32	1.00													
2. Acquisition	0.63	0.24	0.04	1.00												
3. Prior ties	0.10	0.30	0.03	0.00	1.00											
4. Technology relatedness	0.13	0.30	0.06	0.11	0.08	1.00										
5. Indirect ties	0.29	0.56	0.03	-0.02	0.18*	0.18*	1.00									
6. IAP VC backed	0.08	0.27	-0.14	-0.07	-0.10	-0.17*	-0.12	1.00								
7. IAP large pharma	0.23	0.43	0.28*	0.16*	0.26*	0.10	0.31*	-0.15*	1.00							
8. IAP age	52.62	57.38	0.20*	0.21*	-0.04	0.13	-0.03	-0.21	0.11	1.00						
9. IAP alliance experience	13.49	14.01	0.17*	0.06	0.15*	0.23*	0.22*	-0.15	0.54*	0.03	1.00					
10. Acquisition intensity	2.82	2.08	0.01	-0.15	-0.07	0.21*	0.08	0.04	-0.13	-0.06	-0.10	1.00				
11. Acquirer age	23.03	19.71	0.06	-0.15	-0.05	0.23*	0.14	0.12	0.05	-0.02	0.17*	-0.04	1.00			
12. Target preservation	0.74	0.44	0.03	0.15	0.15	0.09	0.12	-0.05	0.19*	0.02	0.23*	-0.18*	0.07	1.00		
13. Cultural distance	6.50	22.78	-0.02	-0.07	0.05	0.11	0.01	-0.03	-0.05	0.13	-0.09	-0.23*	0.31*	0.14	1.00	
14. Geographical dist.	4.92	4.11	0.15	-0.09	0.22*	-0.01	-0.03	-0.02	0.00	-0.10	0.03	-0.18*	-0.09	0.11	0.48	1.00
15. Industry relatedness	2.25	1.73	0.05	0.15*	-0.03	0.15	0.05	-0.10	0.24*	-0.10	0.36*	-0.16	-0.06	0.24*	-0.01	-0.04
Note: N=358. IAP stands for the inherited alliance partner of the acquired firm. * p<0.05																

We now turn to the regression results to examine the effect of acquisition on the acquiring firm's posterior knowledge search behavior towards the inherited alliance network. We begin with the results of the main effects model shown in the column (2) of Table 2. In the main effects models, Hypotheses 1 is supported as the *acquisition* variable is positive and significant with $p < 0.05$. This shows that the acquiring firm indeed cites and refers to more patents of its inherited alliance partners after the acquisition, implying that the acquisition removes the information barrier and instead establishes a knowledge bridge between the acquiring firm and the inherited alliance partners. Due to the nonlinear nature of the negative binomial framework, the coefficient for the *acquisition* dummy is difficult to interpret but the effect size is substantial relative to other explanatory variables and controls. Besides, both *technological relatedness* and *prior ties* are positively and significantly associated with the acquirer's search towards its inherited alliance partner, indicating that they also serve to reduce information asymmetry between the acquirer and its extended alliance network. That is, the more technologically related or if there exists a prior alliance partnership between the acquirer and the inherited alliance partner, then the more frequent the acquirer would cite this partner's patents after the acquisition. However, the main effect of the *indirect ties* variable is insignificant.

We next investigate the interaction effects of *technological relatedness*, *prior ties*, as well as *indirect ties*. While columns (5) shows insignificant results for the indirect ties variable. Both columns (3) and (4) show that there are indeed

substitution effects for technological relatedness and prior ties variables. These findings show that the information intermediating role of the acquisition event can be heavily substituted and weakened if information asymmetry and technology uncertainty is already low or reduced through familiarity of technological areas and prior collaboration experience. It is also sensible that the impact of indirect connections between the two firms are not as stout as the direct relations. Therefore, Hypotheses 2 and 3 are both supported, while hypothesis 5 is inconclusive.

Finally, column (6) shows that all interaction effects are still outstanding and significant in the full model. All three interactions are negatively and significantly associated with the acquirer's knowledge search towards the inherited alliance partner, meaning that all three moderators may serve to substitute the information intermediary role of the acquisition event. Thus, hypotheses 2 and 3 are still supported, and hypothesis 4 is partially supported as well. This potentially implies that the acquisition would provide additional benefits to the acquirer as the acquirer becomes aware of the target firm's alliance network along with an untapped knowledge stock which are beyond the acquirer's search domain and scope.

Additionally, it is also interesting to note that *cultural distance* and *geographic distance* between the acquirer and its inherited alliance partner are both negatively and significantly related to the acquirer's search behavior, and *target preservation* is positively and significantly related to the dependent variable. The negative relationships can potentially be explained due to the high level of information and search costs due to increased distance, and if the target is preserved as a stand-alone

entity, then the acquired firm would better serve to provide the acquiring firm with information of its complete alliance network, in turn better facilitating subsequent knowledge search towards its own knowledge network.

Supplemental analysis

We also seek to investigate the robustness of our results in several ways. First, we assess whether the findings and interpretations are potentially subject to the statistical methods chosen above. In addition to the negative binomial framework, we also run our analysis with a logit framework investigating whether or not there is backward citations. The results of these analyses are consistent with the results shown above in Table 2.

Second, we examine if potential alternative observation windows might affect the results and interpretations. While calculating key variables such as citation counts, technological relatedness, prior ties, and indirect ties, we have removed observations in the year of focal acquisition to reduce noise, and we also calculated these measures using alternative 5-year windows. The results are qualitatively consistent with the results in Table 2.

Table 2-2. Main Regression Results

VARIABLES	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Acquisition X Relatedness			-0.820** (0.358)			-1.224** (0.574)
Acquisition X Prior Ties				-2.257** (0.970)		1.733*** (0.397)
Acquisition X Indirect Ties					-0.131 (0.219)	0.761*** (0.289)
Acquisition		1.061** (0.481)	0.899* (0.491)	1.047** (0.508)	0.646 (0.450)	0.903 (1.094)
Tech Relatedness	0.341* (0.184)	0.726*** (0.236)	1.028*** (0.376)	0.675*** (0.228)	0.720*** (0.234)	0.645 (1.096)
Prior Ties	0.156 (0.428)	0.476* (0.268)	0.313 (0.344)	2.668*** (0.945)	0.477* (0.264)	2.345*** (0.408)
Indirect Ties	0.090 (0.298)	0.185 (0.259)	0.054 (0.118)	0.080 (0.103)	0.199 (0.255)	0.972*** (0.342)
Acquirer Size	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Acquirer Age	0.010*** (0.003)	0.011*** (0.003)	0.008*** (0.003)	0.011*** (0.003)	0.011*** (0.003)	0.017*** (0.006)
Acquirer Acquisition Intensity	-0.023 (0.034)	-0.016 (0.039)	-0.035 (0.034)	-0.016 (0.040)	-0.015 (0.038)	0.011 (0.076)
IAP VC Backing	0.375 (0.229)	0.239 (0.205)	0.428* (0.225)	0.216 (0.198)	0.237 (0.203)	0.606 (0.416)
IAP Alliance Experience	-0.000 (0.008)	0.000 (0.008)	0.002 (0.007)	-0.001 (0.008)	-0.000 (0.008)	-0.000 (0.013)
IAP Age	-0.002 (0.001)	-0.000 (0.001)	-0.002 (0.001)	-0.000 (0.001)	-0.001 (0.001)	-0.001 (0.002)
IAP Large Pharma	-0.018 (0.279)	-0.172 (0.224)	-0.213 (0.259)	-0.077 (0.208)	-0.136 (0.223)	-0.237 (0.329)
Geo Distance	-0.000* (0.000)	- (0.000)	-0.000** (0.000)	- (0.000)	- (0.000)	- (0.000)
Cultural Distance	0.009*** (0.003)	0.010*** (0.003)	0.010*** (0.003)	0.010*** (0.003)	0.010*** (0.003)	0.018*** (0.004)
Industry Relatedness	-0.069 (0.055)	-0.028 (0.048)	-0.052 (0.052)	-0.026 (0.046)	-0.027 (0.047)	-0.013 (0.071)
Target preservation	0.096 (0.218)	0.430** (0.219)	0.121 (0.217)	0.441** (0.201)	0.417* (0.214)	0.890*** (0.331)
Years	Included	Included	Included	Included	Included	Included
Constant	0.723* (0.426)	-1.231* (0.659)	-0.218 (0.650)	-1.167* (0.650)	-0.768 (0.654)	-0.200 (0.894)

N=358. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Third, as shown in Table 3, we run a robustness check using a different method of sample construction. Rather than removing all patents invented by the original target firm inventors from the list of acquiring firm patents after the acquisition, we include these patents of target firm inventors in the sample of acquiring firm patents, and we found that our main results would still hold. As there are potentially more patented co-developed by the original target firm inventors and inventors in the inherited alliance partner firms, we observe that the main effects of the *acquisition* dummy become more stout across all models in Table 3 compared to Table 2. This is potentially because the inherited alliances are still functioning and producing patents after the acquisition. Moreover, the substitution effects of *technological relatedness* and *prior ties* are significant and consistent with our hypotheses, while the interaction effect of *indirect ties* become insignificant. This shows that our results are generally robust and stable.

Table 2-3. Regression with Acquired Firm Inventors

VARIABLES	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Acquisition X Technology relatedness			-1.799* (1.061)			-1.991* (1.116)
Acquisition X Prior ties				-1.883** (0.757)		-2.056*** (0.784)
Acquisition X Indirect ties					-0.629 (1.099)	0.579 (1.115)
Acquisition		1.468*** (0.544)	1.501*** (0.538)	1.715*** (0.524)	1.738* (0.903)	2.472*** (0.645)
Technology relatedness	0.371 (0.700)	0.475 (0.723)	1.233 (1.186)	0.174 (0.646)	0.526 (0.751)	1.773 (1.219)
Prior ties	0.339 (0.356)	0.694 (0.425)	0.731* (0.428)	1.610*** (0.388)	0.657 (0.429)	1.781*** (0.496)
Indirect ties	0.295 (0.198)	0.352* (0.207)	0.334 (0.206)	0.310 (0.232)	0.359* (0.208)	0.280 (0.228)
Acquisition intensity	-0.037 (0.063)	0.022 (0.071)	0.028 (0.070)	0.018 (0.077)	0.033 (0.079)	0.018 (0.084)
Target preservation	0.512 (0.496)	0.341 (0.476)	0.319 (0.472)	0.110 (0.460)	0.340 (0.485)	0.079 (0.441)
Acquirer age	0.024*** (0.008)	0.030*** (0.009)	0.030*** (0.008)	0.026*** (0.010)	0.030*** (0.009)	0.026*** (0.010)
Alliance experience of AP	0.009 (0.014)	0.013 (0.014)	0.014 (0.014)	0.021 (0.014)	0.012 (0.015)	0.023 (0.016)
IAP firm age	0.007*** (0.003)	0.005 (0.003)	0.005 (0.003)	0.003 (0.003)	0.004 (0.003)	0.004 (0.003)
IAP VC backed	-1.95*** (0.611)	-2.14*** (0.605)	-2.16*** (0.561)	-2.07*** (0.570)	-2.16*** (0.618)	-2.06*** (0.532)
IAP large pharma	0.974*** (0.369)	0.875** (0.362)	0.822** (0.374)	0.837** (0.364)	0.940** (0.415)	0.708* (0.461)
Cultural distance	-0.017** (0.009)	-0.016* (0.009)	-0.016* (0.009)	-0.013 (0.010)	-0.015 (0.009)	-0.014 (0.011)
Industry relatedness	0.071 (0.106)	0.009 (0.095)	0.009 (0.097)	-0.006 (0.098)	0.009 (0.097)	-0.005 (0.099)
Years	Included	Included	Included	Included	Included	Included
Constant	7.229*** (0.929)	7.472*** (0.942)	7.436*** (0.961)	7.679*** (0.911)	7.473*** (0.944)	7.608*** (0.929)
N=358. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1						

Discussion

Contributions and implications

We extend the theory of information economics to the literature on knowledge search and advance research by examining the role of the acquisition in facilitating the acquiring firm's post-acquisition search towards the acquired firm's alliance network. Specifically, we examine the information intermediation role of the acquisition in addressing information asymmetries between the acquiring firm and the alliance partners of the acquired firm. Drawing on a quantitative analysis of acquisitions and alliances in the biotechnology sector, we develop hypotheses and test how the acquiring firm may cite patents and search for technology belonging to these inherited alliance partners. In this study, we suggest that the acquiring firm obtains private information from the acquisition target and thus becomes aware of the quality of these inherited alliance partners. Such information enables the acquiring firm to tap into the extend knowledge network of the target firm and better access the technological resources beyond its local reach. Furthermore, these benefits of the acquisition will be greater when the acquiring firm has a high level of information asymmetry and uncertainty about its inherited alliance partner. Conversely, technological relatedness and prior direct or indirect ties between the acquirer and its inherited alliance partner would substitute the information intermediation role of the acquisition target, thus attenuating the acquiring firm's post-acquisition search towards its inherited alliance partners.

Our paper makes several contributions to several literature streams. We bridge together literatures on M&As and strategic alliances to explore how the

acquiring firm searches for knowledge through interorganizational relationships. To begin with, there have been few studies that examine how the acquiring firm inherits alliances after an acquisition, and we extend the scope of M&A research to study not only the acquired firm but also the its extended alliance network. While existing M&A research focused on the benefits derived from within the target firm (e.g. Capron, Dussage, & Mitchell, 1998; Capron, 1999; Ahuja & Katila, 2001; Makri, Hitt, & Lane, 2010), our study reveals that the inherited alliance network can be an important and valuable asset for the acquiring firm, as it provides additional benefits beyond the target firm's own resources and innovation capabilities. Our study therefore adds to the body of research that has examined the importance of boundary-spanning exploration (Dyer & Singh, 1998; Rosenkopf & Nerkar, 2001; Anand, Capron, & Mitchell, 2005) and suggests that the pattern of inherited alliances may become a particularly important means for the acquiring firm to access untapped knowledge domains.

Our study also has implications for research on the theory of information economics. We employ information economics argument to suggest that the acquisition reduces the information barrier between the acquiring firm and its inherited alliance partners. As the acquiring firm gains more information about the scope and quality of the inherited alliance partner's technology, it is more likely to explore technology and cite patents owned by these inherited alliance partners. We argue and show that the acquisition target serves as an information intermediary that significantly reduces the acquiring firm's search costs. In addition, our study

adds to research that investigates alternative mechanisms that would reduce information asymmetries such as technology relatedness (e.g. Mowery et al., 1996; Lane & Lubatkin, 1998; Sampson, 2007), prior ties (e.g. Gulati, 1995; Gulati & Gargiulo, 1999; Schildt & Laamanen, 2006), and indirect ties (e.g. Van de Ven, 1976; Raub & Weesie, 1990; Burt & Knez, 1995). Future research can examine other types of interorganizational ties (e.g. common VCs, common clusters) through which the acquiring firm can learn about its resources and capabilities of its inherited alliance partners (Rosenkopf & Almeida, 2003).

Interestingly, our results also help to advance our knowledge about the post-acquisition integration strategy. Typically, acquisition scholars have focused their attentions on the post-acquisition firm performance (e.g. Chaudhuri & Tabrizi, 1999; Cording et al., 2008; Datta, 1991; Paruchuri et al., 2006). To the best of our knowledge, previous acquisition research rarely goes beyond the direct consequences of acquisitions for the acquiring firm. Our paper shows that the post-acquisition integration strategy may have broader consequences such as interorganizational knowledge search and exploration. Future research can examine the dynamics of the integration strategy and the inherited alliance partners.

Limitations and future research directions

In addition to the research possibilities already noted, extensions might also pursue additional research opportunities presented by some limitations of this study. First, we test our hypotheses using patent data in the biotechnology industry, which can only track the exploration patterns of innovations that are successful enough to

have resulted in patents. As firms certainly undertake knowledge search that do not results in granted patents, it would therefore be valuable in future research to measure knowledge search in more detailed and dynamic ways.

Second, a technology venture's search behavior may be fundamentally different due to its internal structure of R&D, and it would be interesting to examine how the internal organization of R&D would influence the focal firm's external search pattern. While our study currently explores the relationship of organizational boundary-spanning and knowledge search, we consider the organization as a single entity and information flows "freely" across different research units and divisions within the organization. However, as we have shown that target integration does play a role in knowledge absorption and recombination, future research can examine how the internal structure of an organization (Argyres and Silverman, 2004; Arora, Belenzon, and Rios, 2014) might further affect its knowledge search trajectory.

Third, our study focuses on the organizational boundary spanning via M&As, and future research can explore other types of organizational boundary spanning via other corporate actions. For example, research have shown that geographical agglomeration and geographical co-location are associated with alliance formation and partnerships (McCann, Reuer, and Lahiri, 2016; Ryu, McCann, and Reuer, 2018). Following similar arguments as above, one avenue for further research is to examine how an organization's knowledge search direction is likely to be affected by the clustering or the geographical distance from its partners.

Conclusion

Extant literature mostly focuses on accessing the acquired firm's resources and capabilities, while ignoring the resources and capabilities from the extended network of the acquired firm. However, knowledge from the extended network is important as it enables the firm to search beyond its organizational boundaries and knowledge scope. Few studies have analyzed the acquiring firm's post-acquisition search behaviors related to the acquired firm's alliance partners. In this study, we show that the acquirer is more likely to direct its knowledge search towards the acquired firm's alliance partners as the acquirer gains more information about them through the information mediation of the acquisition target. Our paper suggests that acquisitions may provide the acquirers with access to not only the acquired firm's knowledge base, but also that of the acquired firm's alliances partners. We hope that this study encourages more research on knowledge search in the inherited alliances context as well as work joining the literatures on information economics.

CHAPTER 3.

Organizational Structure Matters: An Analysis of R&D Centralization and Knowledge Search in Strategic Alliances

Introduction

To achieve competitive advantage in the high-tech industry often requires firms to constantly search for new knowledge and innovate (Nonaka and Taeuchi, 1995; Simonin, 1997). As firms face path dependencies and local search constraints (Dierickx and Cool, 1989; Leonard-Barton, 1995), it can be extremely difficult to generate knowledge on its own, and firms often engage in knowledge exploration beyond their organizational and technological boundaries (Rosenkopf and Nerkar, 2001; Nelson and Winter, 1982; March and Simon, 1958). Therefore, forming strategic alliances has become common organization practice in many industries such as the bio-pharmaceutical industry. Organizations constantly engage in alliance activities largely to acquire external knowledge that is difficult to develop internally (Rosenkopf and Almeida, 2003). By forming an R&D alliance, firms can leverage on the technological capabilities of their partners. Strategic alliances with joint research and development often incur a high level of exchange of know-how and technology transfer (Mowery, Oxley, and Silverman 1996), and are often subject to hazards such as undesirable knowledge spillovers or misallocation of research effort (Kale, Perlmutter, and Singh 2000; Lerner and Malmendier 2010).

Therefore, the issue of knowledge search and sourcing strategy has become increasingly important in the discussion of alliance research and cooperative strategy. A substantial body of work has advanced our understanding of the knowledge exchange between alliance partners (e.g. Hamel et al. 1989; Doz, 1996; Dyer, 1997; Reuer & Arino, 2007). Given that the R&D generation process is mostly intangible, a question that remains to be further explored is how firms can efficiently

perform knowledge search and evaluations to benefits from alliance partner's technological assets and capabilities (Rangan, 2000; Vanhaverbeke et al., 2002; Villalonga & McGahan, 2005).

An assumption implicitly made in traditional alliance research is to treat each partner firm as a single entity. Once the accessed knowledge from the alliance partners has successfully been transferred to the focal firm, it can 'freely' be recombined with firms' existing knowledge base and with other alliances. By doing so, effects of the firm's internal structure are often overlooked. To date, only a small number of empirical studies have paid attention to firms' internal management of knowledge, particularly the centralization of R&D, and how it may affect knowledge creation and recombination through external partnerships (Argyres and Silverman, 2004; Zhang, Baden-Fuller, and Mangematin, 2007; Arora, Belenzon, and Rios, 2014). Using research budgetary data, Argyres and Silverman (2004) find that firms that operate a centralized R&D will generate innovations with larger and broader impact on subsequent technological evolution, relative to decentralized firms, and they also link R&D centralization to different types of innovation and show that centralized R&D will facilitate distant knowledge search. Furthering this stream of research, Ashish et al. (2014) measure R&D centralization with the patent assignment patterns data and they find that centralized firms tend to engage in less M&As for knowledge generation, but rather they patent more technologies in house (potentially using technologies learned through external collaborations). In addition, Zhang et al. (2007) link R&D centrality to alliance research, and show that R&D centrality

positively influence the firm's propensity to form strategic alliances, as the centralized R&D may benefit more from alliances compared to the decentralized R&D. Following the abovementioned prior work, we aim to extend our current understanding of R&D centralization, and how it relates to the external knowledge acquisition, specifically through technological search from alliance partners.

In this paper, we set out to examine whether and how the internal structure of an organization may play a role in disrupting the knowledge diffusion process related to its alliance activities. We define R&D centralization based on the presence of an R&D head on the executive team, using data regarding the authoritative reporting and hierarchy (Lerner and Wulf, 2007). In a centralized firm, the presence of the R&D head in the executive team can use its R&D authority to facilitate non-specific, spillover-heavy projects, which decentralized firms do not have incentives to do. If a client firm has a decentralized R&D structure, internal boundaries between business units may impede information exchange and knowledge transfer within the firm and decrease the likelihood of successful knowledge recombination.

Compared to the firm with a decentralized R&D structure, a centralized client firm is more aware about the knowledge that resides in different strategic business units (SBUs) within the firm, as well as the extended knowledge portfolio of their alliance partners. In this way, it can use and recombine knowledge acquired from different partners and diffuse it more 'freely' within the organization. Consequently, we submit that a client firm with a centralized R&D structure is more likely to extend knowledge search more heavily towards its alliance partners.

Also, centralized firms are more likely to capture high-level innovation and knowledge with broad scope, so that they can use that knowledge in their own research development process in the future. In turn, we submit that when R&D coordination is at issue, the centralized R&D structure can provide superior efficiency, and for this reason, the centralized R&D structure is also associated with knowledge search with broader search breadth in R&D alliances. In addition, a client firm with a centralized R&D structure tends to have a clear reporting channel and a direct decision-making structure, which would help lower information and coordination costs within the client firm. For this reason, a centralized client firm is also more likely to diffuse the externally sourced technical know-how into a wider technological domain within the client firm.

We develop and test our hypotheses on a dyadic sample of 960 non-equity R&D bio-pharmaceutical alliances. We explore the relationship between centralized R&D structure of the pharmaceutical firms and their external knowledge sourcing strategies their alliance partners. Our empirical context is the bio-pharmaceutical industry, in which pharmaceutical firms (referred to as the client firms) often form alliances with relatively small biotech firms (referred to as R&D firms). Due to the fact that around 86% (source: ReCap) alliances signed in the bio-pharmaceutical industry are not equity-based, we focus our study on non-equity alliances from the client firm's perspective, and examine how the internal structure of the client firms exert impact on the client firm's knowledge search strategy related to its alliance partners. From our analyses of the 960 non-equity alliances for the time period 1990-

2011, we find that firms that have centralized R&D structure conduct knowledge search more heavily from their alliance partners, that they tend to conduct broader knowledge search by citing a wider range of technological classes from their partners, and that they use such cited knowledge in a wider range of therapeutic areas.

We contribute to the organizational structure research and bridge it with research on organizational boundary-spanning, strategic alliances, and knowledge search. First, our paper complements existing studies aimed at analyzing the link between R&D centralization and external knowledge acquisition. Firms typically achieve external knowledge acquisition through M&As and strategic alliances (Rosenkopf and Almeida, 2003), and it is often assumed that firms would easily combine such acquired knowledge with its existing knowledge stock. However, as we consider the organization's internal knowledge repository and that knowledge is often segregated across different research units and teams ((Daneels, 2011)), this study extends our understanding of knowledge acquisition and recombination within the organizational structure.

Second, we offer insights on how centralization of R&D within a firm can exert impact on the client firm's knowledge search behavior towards its strategic alliance partners. By considering how pharmaceutical firms manage R&D and capabilities internally across research units, we find that R&D centralization plays an important role in shaping the client firm's external knowledge search trajectory towards to its alliance partners (Arora and Gambardella, 1994; Zheng et al., 2007). Drawing upon both alliance and patent data, our results show that firms with centralized R&D are

focused on higher level innovation and thus are more likely to diffuse and recombine external alliance partner knowledge more effectively within their own research units.

Third, as empirical studies on organizational structures are rare, we provide empirical evidence using data in bio-pharmaceutical industry to proxy for the R&D structure of the firm. We measure R&D centralization based on whether the position of an R&D head is present in the top management team (Lerner and Wulf, 2007). Our findings are also in line with prior research on R&D centralization using alternative measures (Argyres and Silverman, 2004; Arora et al., 2014), demonstrating that firms with centralized R&D are associated with better communication efficiency when it comes to R&D coordination and are more likely to generate innovations with a wider breadth of technological classes and therapeutic areas.

Hypotheses Development

Strategic Alliances and External Knowledge Acquisition

A firm's critical resources, however, may span beyond firm boundaries (Dyer & Singh, 1998). Since a firm's knowledge stock is accumulated through interactions among diverse external organizations, a breakthrough innovation is hardly achieved solely based on the knowledge within a single firm (Badaracco, 1991). Organizations depend on the inter-organizational cooperation to access resources that are essential to achieve their goals but under the control of other organizations (Gulati & Gargiulo, 1999; Pfeffer & Salancik, 1978). Recent empirical studies have shown the contribution of R&D alliances to the firm's knowledge base, knowledge recombination, and innovation performance (Baum, Calabrese, & Silverman, 2000;

Ahuja, 2000; Rothaermel & Hess, 2007). Scholars state that innovation process should be understood as a comprehensive mechanism which includes the focal firm's relationship with other organizations through which the firm can access external resources.

Strategic alliances are long-term contracts between legally distinct organizations that provide for sharing the costs and benefits of a mutually beneficial activity (Robinson, 2008). Research on strategic alliances has blossomed since the 1980s (Glasmeier, 1991; Hagedoorn, 1993), with one essential goal aiming to understand how firms manage and gain technical capabilities from alliance activities (Ireland et al., 2002). To gain a more detailed understanding of the relationship between the internal organizational structure of R&D and knowledge search, we examine how bio-pharmaceutical firms search, cite, and utilize technologies from their alliance partners.

Organizational Structure and R&D Centralization

In technology intensive industries, in which R&D matters the most, a firm can have a dedicated R&D team that is supervised by an R&D head, who holds a seat in the executive team of that firm. Alternatively, the firm can have different R&D teams focused on different technologies. When the firm appoints a dedicated R&D head, we consider it as a centralized R&D structure. In this set-up, the decisions regarding the R&D management comes through the dedicated R&D head. Naturally, researchers have found a trade-off between these forms of R&D structure. In the centralized structure, there is a single executive in charge of the firm's research activities, who is typically titled as the R&D head and reports directly to a corporate-level executive

such as the CEO or President. In the decentralized structure, research is conducted exclusively within divisions or business units, and R&D directors report to divisional general managers instead. In general, the centralized R&D structure provides a proper communication channel and facilitates the flow of information. Centralized research would favor knowledge search in solving broader, non-specific research challenges, while decentralized research will favor investment in solving narrower, business-unit specific research challenges (Argyres and Silverman, 2004). If corporate-level management refrains from delegating R&D management authority to the divisions, it can use this authority to delegate non-specific, spillover-heavy R&D projects that the divisions do not have incentives to fund themselves. Thus, the client firm with a centralized structure is more likely to coordinate the externally acquired technical know-how in such a way that the client could capture a wider range of benefits through various ways of knowledge recombination across different research units within the firm.

We develop hypotheses below that a client firm with a centralized R&D is associated with knowledge exploration from its alliance partner with a great search volume, a wider breadth, as well as a wider technology application domain when it comes to using the external technical know-how in generating its own innovations. Scholars have found that firms with a centralized R&D structure tend to be innovation-focused and are more likely to pursue basic and ground-breaking research, it would seem more likely that a centralized R&D structure leads to a serendipitous discovery. We claim that the internal R&D structure of a firm plays a

major role in the distribution of rights during signing the alliance contracts. Firms with centralized R&D structure invest more in R&D, generate more patents and create innovations with larger and broader impact on subsequent technological innovation (Argyres & Silverman, 2004; Arora et al., 2014). These firms, with a central R&D head in the executive team, will focus more on gaining long term competitive advantage. Also, the client firm with a centralized structure will have much more awareness about different activities in the R&D division. Thus, knowledge recombination will be much easier for these types of firms, and centralized firms will be more likely to capture the upstream rights from its alliances so that they can use that knowledge in their other projects and recombine knowledge. Therefore, we propose that firms with centralized R&D structure are more likely to conduct knowledge search in alliance contracts. We hypothesize that client firms with centralized R&D structures tend to cite innovation and technologies from their alliance partners with a greater search volume.

H1. A client firm with a centralized R&D is associated with a greater search volume, when conducting knowledge search from its alliance partner.

Second, we argue that a client firm with a centralized R&D is more like to engage in knowledge exploration from the alliance partner with a greater breadth. Specifically, a centralized firm plays a major role in the coordination of alliance activities and the exchange of know-how within the firm. Firms with a centralized R&D structure are more capable of reducing the internal transaction costs associated with the coordination of activities across business units. Centralized firms achieve a

clear chain of communication, thereby reducing information loss and tightening control over information flow (Argyres and Silverman, 2004). Thus, such firms are better able to coordinate among multiple activities in an alliance more effectively than firms relative to firms with a decentralized R&D structure. Consequently, when forming an alliance, centralized firms are more likely to capture a broader range of R&D activities in an alliance in order to benefit from potential knowledge recombination.

H2. A client firm with a centralized R&D is associated with a wider search breadth, when conducting knowledge search from its alliance partner.

Third, we argue that the client firm's internal structure of R&D plays an important role in determining the range of technologies cited from the alliance partner. Prior studies address the issue of the misappropriation hazards for alliances that have high interdependence. As the level of interdependence between alliance partners increases, partners are obligated to work closer with each other and the chance of knowledge leakage and misappropriation rises (Aiken & Hage, 1968; Thompson, 1967; Van De Ven, Delbecq, & Koenig, 1976). When the client firm has an executive who serves as the R&D head and oversees all R&D activities of the firm, the R&D head can access and integrate the technical know-how from different business units more efficiently relative to a client firm without a centralized R&D head. The R&D head can serve as a sole decision maker for R&D activities, thereby providing a more efficient channel of information flow in a clear hierarchical order. As the amount of task interdependence between alliance partners increases the need

for effective coordination and better capability of knowledge recombination, a centralized firm can afford to cite more broadly from an alliance partner.

H3. A client firm with a centralized R&D is associated with patenting in a wider technology domain, when conducting knowledge search from its alliance partner.

Method and Analysis

Data and Sample

The rapid technological breakthrough in global bio-pharmaceutical industry gives us an appropriate setting to test our hypotheses. There are several reasons why we select this industry as the setting for our study. First, the bio-pharma industry has been constantly undergoing radical innovation (Hagedoorn, 1993). The biotech firms face continuous adaptation pressure, and in turn frequent alliances with large client firms are necessary and essential for their survival. This makes it an ideal context to analyze the technological innovation behavior of incumbent pharmaceutical firms (Katila and Ahuja, 2002). Second, in this industry, R&D activities are dispersed among large pharmaceutical firms (typically serving as client firm) and new biotechnology firms (typically serving as R&D firm), and thus high levels of innovation and knowledge recombination often occur through alliance activities (Powell et al., 1996). Although many studies have studied the impact of alliance on innovation performance in this industry (e.g., Rothaermel, 2001; Rothaermel and Deeds, 2004), rarely have they examined how the internal structure of R&D impacts the knowledge search and diffusion from alliance partners into the

client firms. For the purpose of this study, we only consider dyadic alliances and leave out the multilateral alliances for future research.

Our sample includes 960 non-equity based dyadic alliances formed between US firms during the years from 1990 to 2011. We construct our sample from the alliances that are available in Recap, which is a comprehensive database covering the inter-organizational agreements in the bio-pharma industry (Schilling, 2009). We gather information on 960 dyadic alliances that has non-equity based alliance structure. Then we collect information regarding the R&D structure of the client firms engaged in these alliances. To get this information, we use the annual reports of the firms and complemented it with the web-archival data source (<https://archive.org/web/>). To identify whether the client firm has a centralized R&D structure when the alliance is formed, we collect information about whether an R&D head position is present on the executive team for the previous year from the formation year of the alliance.

To reduce noise and deal with missing information in the data source, we exclude contracts that are signed before 1990, because a) alliance activities drastically increased after 1990 and b) it is difficult to get the R&D centralization data for the firms prior 1990. We further measure the set of control variables using patent assignments and citations data from USPTO, PatentsView by Google, as well as information from VenturExpert by ThompsonOne. Finally, we reach a dyadic sample of 960 alliances formed by 597 firms.

Dependent Variables

We use a combination of patent data from USPTO, PatentsView by Google, and the R&D alliance data from ReCap (Schilling, 2009) to calculate our dyad-year level dependent variables. To better understand the knowledge search diffusion process through alliance partners, we create three different dependent variables to capture the *search volume*, *search breadth*, and *search domain* that the focal client firm is engaging with an alliance partner in the focal year. We elaborate on the calculation of the three dependent variables below.

Search volume from the alliance partner. For each dyad-year observation, we measure the client firm's search volume from the focal alliance partner as the number of backward citations accumulated in the 5 years from the focal year (Rosenkopf & Nerkar, 2001; Katila & Ahuja, 2002; Rosenkopf & Almeida, 2003). To obtain this measure, we start from collecting and matching patent citation and assignee data where a client firm patent is citing an R&D firm alliance partner patent by year. This enables us to track patent information based on assignee name and year information. Coupling with patent citation data and alliance dyads data, we match and select the client firm's patents where the focal R&D partner firm's patent is cited. Then we create a list of observations in the format of client-patent-citing-R&D-patent-year, and we aggregate them by creating a 5-year rolling window from the focal year. We then calculated the cumulative sum of backward citations that any of the client firm's patents has cited any of the R&D partner firm's patents within the 5-year rolling window. Thus, this measure indicates the extent to which the client firm searches specific technological innovation from the focal alliance partner. We expect that the

client firm's search volume from the focal alliance partner is a proxy for technological capabilities that the client firm might benefit from the alliance. Relating this variable to Hypothesis 1, we expect that a centralized client firm is more likely to generate higher and broader impact innovations through external learning and collaborating from its alliance partner. Thus, we expect that R&D centralization is associated with a larger search volume from the alliance partner.

Search breadth from the alliance partner. For each dyad-year observation, we measure the client firm's search breadth from the focal alliance partner as the number of unique technological classes that the client firm is citing from the R&D alliance partner. To specify, this measure refers to the unique number of technological classes of the alliance partner. To calculate this variable, we again start with a list of the client-partner firm dyads where any of the client firm's patents is citing any of the R&D alliance partner firm's patents within the 5-year window from the focal year. Then we calculate the cumulative sum of the unique numbers of IPC technological classes of the R&D alliance partner firm which the client firm has cited within the 5-year rolling window from the focal year. Therefore, this measure refers to the breadth of the client firm's technological search breadth in relation to its focal R&D alliance partner. A high value in the search breadth potentially indicates that the client firm is gaining from a wide range of technological or therapeutic areas from the focal R&D alliance partner. Relating this variable to Hypothesis 2, we expect that the client firm with a centralized R&D is better capable of evaluating and recombining external knowledge, thus associated with a wider search breadth from the alliance partner.

Search domain from the alliance partner. For each dyad-year observation, we measure the client firm's search domain from the focal alliance partner as the number of unique technological classes which the client firm is patenting under, after citing from the R&D alliance partner. To specify, this measure refers to the unique number of technological classes of the client firm. To calculate this variable, we again start with a list of the client-partner firm dyads where any of the client firm's patents is citing any of the R&D alliance partner firm's patents within the 5-year window from the focal year. Then we calculate the cumulative sum of the unique numbers of technological classes of the client firm under which the client firm has patented its own innovation within the 5-year rolling window from the focal year. Therefore, this measure refers to the client firm's technological domain that has benefited its focal R&D alliance partner. A high value in the search domain potentially indicates that the client firm is able to gain benefit from the focal R&D alliance partner and therefore generate innovation in a wide range of technological or therapeutic areas. Relating this variable to Hypothesis 3, we expect that the client firm with a centralized R&D is better capable of using external knowledge and generating innovations in a wide technological domain, thus associated with a wider search domain from the alliance partner.

To briefly sum up the above dependent variables, it is worth noting that the first dependent variable, *search volume*, is a count of backward citations. The second dependent variable, *search breadth*, is a count of unique technological classes of the

R&D firm alliance partner, whereas the third dependent variable, *search domain*, is a count of unique technological classes of the client firm *per se*.

Independent Variable

We follow the abovementioned definition to calculate our measure for *R&D centralization*. We identify whether the client firm has a centralized R&D structure when the alliance is formed. We use the annual reports of the firms and complemented it with the web-archival data source (<https://archive.org/web/>) to collect information about the client firm's executive team set-up for the previous year from the focal year. Our variable for R&D centralization takes the value of 1 if the firm has a R&D head in its executive team. That is, the client firm is considered to operate a centralized R&D if all the R&D decisions are made at the corporate level by the R&D head in the executive team. If the firm does not have a R&D head in the executive team, then we code the variable as 0. It is worth noting that we expect this measure to be relatively stable over an extended period of time. While the stability of this measure might dampen the significance of our empirical analysis results, it resolves the reverse causality issue where the R&D centralization is the result of alliance activities. We do not claim to test for the causal relationship between the R&D centralization and the client firm's alliance partner knowledge search, yet we expect that our empirical finding shows that centralization exerts an impact on how the client firm searches external knowledge from its alliance partners.

Control variables

At the alliance dyadic level, if the alliance partners have overlapping technological expertise, then the client firm is expected to have an enhanced

familiarity and technical understanding of the technology, and thus would likely lead to an in-depth search and learning from the alliance partner. Therefore, we control for *technology overlap* between the client firm and R&D firm alliance partner. We calculate the technology overlap measure using the cosine similarity method (Jaffe, 1986; Sampson, 2007). We examine the extent to which client and R&D firm have patents in same technology class. This gives us a measure that captures the relative position of client and R&D firm in terms of technology. We use the patent classes assigned by USPTO (patent data source: PatentsView) and calculate the technology similarity between the partner firms. We first generate the each firm's patent portfolio (per year) and subsequently add the patent class to this distribution. This vector looks like: $F_i = (F_i^1 \dots F_i^s)$, where F_i^s represents the number of patents assigned to partner firm i in patent class s . Then,

$$Technology\ overlap = \frac{F_i F_j'}{\sqrt{(F_i F_i')(F_j F_j')}} \quad \text{Where } i \neq j.$$

This measure varies from 0 to 1, with a value of 0 indicating the greatest possible technological distance between partner firms (Sampson, 2007).

Having a prior relationship may also affect how the partners decide the scope and interdependence of their future alliances. Having a prior tie might indicate that the client firm has previous engaged in alliance activities with the same alliance partner. Therefore, we control for *prior ties* between the alliance partners. This variable takes value 1 if the partners had an alliance in prior years else it takes the value 0. While a pleasant prior relationship would shape a more in-depth knowledge search trajectory, the existence of prior ties may not indicate a positive relationship

between the partners. Thus, we control for prior ties without expecting a certain outcome.

Also, alliance characteristics may vary depending upon the stage of the development process. This leads to firms having different expectations from alliances that are more mature in terms of the development process than a nascent one. Therefore, we control for the whether the alliance is at the *preclinical alliance* stage. We created an indicator variable which takes a value of 1 if the alliance was signed at the early stage (shown as *preclinical alliance*) and takes a value of 0 if the alliance was signed at a later stage (phase I, phase II or higher clinical trial) (Folta, 1998).

At the client firm and the R&D firm alliance partner level, prior *alliance experiences* of partner firms might impact the characteristics of future alliances. Therefore, we control for the experiences of both the client firm and the R&D firm (focal alliance partner). We calculate the alliance experience of a firm by counting the total number of alliances that firm has been involved in the past. We expect that both the *alliance experiences* of both firms would have a positive effect on the knowledge search efficiency, thus positively associated with the search volume, search breadth, as well as the search domain.

We also control for the existing *knowledge stock* of the both the client firm and the R&D firm. We calculate the *knowledge stock* as the total number of unique technological classes a firm has patented in the prior years of the alliance. We expect that a large existing knowledge stock of the client firm would play a substitutive role for R&D centralization, thus resulting in higher- and broader- impact know search.

Similarly, we also expect that a large existing knowledge stock of the R&D firm alliance partner is also associated with higher values in *search volume*, *search breadth*, and *search domain*, as it would create high incentives for the client firm to gain technological capabilities from the R&D firm. We calculate this measure for a 3-year, 5-year and 7-year windows, respectively. We find consistent results are consistent with all year windows.

For the similar reason, we expect that an R&D firm with high quality innovations is associated with high incentives for the client firm to conduct knowledge search with. Therefore, we control for the *R&D firm technology value* as the number of forward citations in the 5-year rolling window prior to the focal year, in order to capture the R&D firm quality and the client firm's search incentive accordingly.

In addition, we control for *years* and *therapeutic areas* of the alliances in our sample, in order to account for certain degree of noise due to macroeconomic conditions such as time trends and industry characteristics.

Results

Statistical analysis

Due to the count nature and over dispersion of patent citations data, we employ a negative binomial framework to test our hypotheses. Table 1 reports the descriptive statistics, and Table 2-4 show respectively the regression results for our hypotheses on the client firm's knowledge *search volume*, *search breadth*, and *search domain* from its alliance partners. Our final sample consists of 960 dyad-year observations in our sample with a reasonable amount of variations occurring across the observation window. At a given year, the maximum number of *search volume* from the alliance

partner in our sample is 23, and the minimum is 0. At a given year, the maximum number of *search breadth* from the alliance partner in our sample is 21, and the minimum is 0. At a given year, the maximum number of *search domain* from the alliance partner in our sample is 16, and the minimum is 0. As there are some zero values occurring in the sample across the dependent variables, we conducted an alternatively zero-inflated framework to test our results, which are qualitatively similar to the main results shown in Tables 2-4. It is also worth noting that the correlations between the three dependent variables are high, which is reasonable. However, the variation inflation factor (VIF) across the regression models are well under the cut-off level (10) for any given variable (Neter, Kutner, Nachtsheim, & Wasseman, 1996). The average VIFs across the reported models are below 4, and the maximum VIF scores are less than 5. We report the mean and variance for *R&D centralization* to be 0.77 and 0.42, indicating a slightly higher portion of centralized firms in our sample with respectable variations. In general, robust standard errors are shown in the main results, and we find support for all three hypotheses. Next, we discuss the main results of Tables 2-4 below.

Table 3-1. Descriptive Statistics

Variable	Mean	Std. Dev.	1	2	3	4	5	6	7	8	9	10	11
1. Search volume	2.38	15.85	1										
2. Search breadth	0.62	2.01	0.63										
3. Search domain	0.45	1.58	0.60	0.77									
4. Centralization	0.77	0.42	-0.01	-0.07	-0.03								
5. Technology overlap	0.32	0.32	0.14	0.29	0.28	-0.07							
6. R&D technology value	4.61	3.71	0.14	0.24	0.28	-0.07	0.15						
7. Prior ties	0.34	0.93	0.10	0.11	0.13	0.05	0.22	0.18					
8. Preclinical alliance	0.08	0.28	-0.01	0.01	-0.01	-0.02	0.09	-0.06	-0.01				
9. Client alliance experience	40.85	40.50	-0.02	-0.07	-0.09	0.33	-0.02	-0.13	0.08	0.10			
10. R&D firm alliance experience	8.25	11.39	0.17	0.25	0.22	-0.03	0.26	0.45	0.43	-0.01	0.02		
11. Client knowledge stock	52.88	40.39	0.06	0.12	0.10	0.11	0.06	-0.01	0.07	-0.01	0.26	0.10	
12. R&D firm knowledge stock	12.15	17.90	0.16	0.35	0.36	-0.06	0.34	0.64	0.31	-0.01	-0.10	0.60	0.01
Note: N = 960.													

Table 2 shows the negative binomial regression results for our first dependent variable, *search volume from the alliance partner*. As Table 2 shows, model (1) is the baseline model with just control variables, and model (2) shows the main effect with the explanatory variable *R&D centralization*. The main effect of *R&D centralization* is 0.657 ($p < 0.01$), indicating that *R&D centralization* is positively associated with the knowledge search volume from the client firm's alliance partner. So, hypothesis 1 is supported. That is, a client firm with a centralized R&D is more likely to engage in knowledge search more extensively and heavily from its alliance partner. This implicitly implies that the centralized R&D structure of the client firm is associated with a greater incentive to search and recombine external knowledge through collaboration in order to generate high- and broad-impact innovations.

It would also be interesting to check the results of some control variables in the models. As expected, *technology overlap* is positively and significantly associated with *search volume* ($p < 0.01$), indicating that the client firm is more likely to conduct extensive and in-depth search from its alliance partner if the client firm is in the same technology space with the alliance partner. Also, both *R&D firm technology value* and *R&D firm knowledge stock* are positive and significant, indicating that the client firm is more incentivized to search and cite from the R&D firm alliance partner if the partner has demonstrated its technology quality and breadth. In addition, both *client firm alliance experience* and *R&D firm alliance experience* do not matter as much in the client firm's knowledge search.

Table 3-2. Regression on Client Firm's Search Volume from Alliance Partners

VARIABLES	(1) CLIENT CITES RD	(2) CLIENT CITES RD
Centralization		0.657** (0.291)
Client knowledge stock	0.012*** (0.003)	0.012*** (0.003)
Client alliance experience	-0.001 (0.005)	-0.005 (0.005)
Technology Overlap	1.250*** (0.183)	1.313*** (0.180)
Prior ties	0.190* (0.104)	0.174* (0.099)
Preclinical alliance	0.633 (0.405)	0.598 (0.384)
R&D firm technology value	0.247*** (0.061)	0.247*** (0.061)
R&D firm alliance experience	0.011 (0.012)	0.011 (0.011)
R&D firm knowledge stock	0.042*** (0.009)	0.043*** (0.010)
Constant	-3.868*** (0.776)	-4.031*** (0.817)
Years FE	102.17***	109.46***
Therapeutic areas	YES	YES
alpha	10.06	9.93
χ^2	2013.85	1902.54

N = 960. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

It is worth noting that the alpha estimates for the dispersion parameters are significantly positive (10.06 and 9.93), indicating that the negative binomial framework is better fitted than a Poisson model. Also, the estimates for χ^2 are significantly great (2013.85 and 1902.54), showing the overall model fitness.

Turning to hypothesis 2, Table 3 shows the negative binomial regression results for our second dependent variable, *search breadth from the alliance partner*. As Table 3 shows, model (3) is the baseline model with just control variables, and model (4) shows the main effect with the explanatory variable *R&D centralization*. The main effect of *R&D centralization* is 0.401 ($p < 0.1$), indicating that *R&D centralization* is positively yet weakly associated with the knowledge search breadth from the client firm's alliance partner. So, hypothesis 2 is supported. That is, a client firm with a centralized R&D is likely to engage in knowledge search more widely from its alliance partner. This moderately implies that the centralized R&D structure of the client firm is associated with a greater incentive to search and recombine external knowledge through collaboration in order to generate broader impact innovations. One possible explanation for the relatively weak effect on the client firm's search breadth is that the client firm is likely to achieve search breadth from its alliance portfolio instead of just one alliance partner.

Table 3-3. Regression on Client Firm's Search Breadth from Alliance Partners

VARIABLES	(3) CLIENT CITES RD	(4) CLIENT CITES RD
Centralization		0.401* (0.236)
Client alliance experience	-0.003 (0.004)	-0.001 (0.004)
Technology Overlap	0.928*** (0.128)	0.999*** (0.152)
Prior ties	0.053 (0.111)	-0.010 (0.189)
Preclinical alliance	0.644 (0.396)	-0.041 (0.429)
R&D firm technology value	0.665*** (0.213)	1.039*** (0.192)
R&D firm alliance experience	0.004 (0.009)	0.024*** (0.009)
Constant	-2.483*** (0.682)	-1.792*** (0.605)
Years FE	61.59***	65.60***
Therapeutic areas	YES	YES
alpha	4.67	5.91
χ^2	6585.78	10171.69
N= 960. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1		

It would also be interesting to check the results of some control variables in the models (3) and (4). As expected, *technology overlap* is positively and significantly associated with *search volume* ($p < 0.01$), indicating that the client firm is more likely to conduct comprehensive knowledge search from its alliance partner if the client firm is in the same technology space with the alliance partner. Also, *R&D firm technology value* is positive and significant, indicating that the client firm is more incentivized to search and cite broadly from this R&D firm alliance partner if the partner has demonstrated its technology quality. Again, both *client firm alliance experience* and *R&D firm alliance experience* do not matter as much in the client firm's knowledge search breadth.

It is again worth noting that the alpha estimates for the dispersion parameters are significantly positive (4.67 and 5.91), indicating that the negative binomial framework is better fitted than a Poisson model. Also, the estimates for χ^2 are significantly great (6585.78 and 10171.69), showing the overall model fitness.

Finally, Table 4 shows the negative binomial regression results for our third dependent variable, *search domain from the alliance partner*. As Table 2 shows, model (5) is the baseline model with just control variables, and model (6) shows the main effect with the explanatory variable *R&D centralization*. The main effect of *R&D centralization* is 0.466 ($p < 0.05$), indicating that *R&D centralization* is positively associated with the knowledge search domain from the client firm's alliance partner. So, hypothesis 3 is supported. That is, a client firm with a centralized R&D is more likely to search and cite from its alliance partner and use these technologies in a

wider technological domain in its own innovations. This implicitly implies that the centralized R&D structure of the client firm is associated with a greater ability to search and recombine external knowledge through collaboration, in order to generate architectural innovations in untapped domains.

It would also be interesting to check the results of some control variables in the models. As expected, *technology overlap* is positively and significantly associated with *search volume* ($p < 0.01$), indicating that the client firm is more likely to conduct extensive and in-depth search from its alliance partner if the client firm is in the same technology space with the alliance partner. Also, both *R&D firm technology value* and *R&D firm knowledge stock* are positive and significant ($p < 0.01$), indicating that the client firm is more incentivized to use the cited technologies and innovate in its own technical domain. In addition, *preclinical alliance* is positive and significant in models (5) and (6), implying that the client firm is more likely to cite from R&D firm in its early stage alliance and use the technology in a wider domain. This effect potentially speaks to the features of early stage alliances where early stage technologies are more likely to be recombined and patented in untapped technology domains.

It is again worth noting that the alpha estimates for the dispersion parameters are significantly positive (3.71 and 3.04), indicating that the negative binomial framework is better fitted than a Poisson model. Also, the estimates for χ^2 are significantly great (9723.34 and 12468.60), showing the overall model fitness.

Table 3-4. Regression on Client Firm's Search Domain from Alliance Partners

VARIABLES	(5) CLIENT CITES RD	(6) CLIENT CITES RD
Centralization		0.466** (0.203)
Client knowledge stock	0.013*** (0.004)	0.012*** (0.004)
Client alliance experience	-0.004 (0.004)	-0.002 (0.004)
Technology Overlap	0.906*** (0.125)	0.911*** (0.151)
Prior ties	0.042 (0.115)	0.014 (0.209)
Preclinical alliance	1.253*** (0.401)	0.881* (0.451)
R&D firm technology value	1.117*** (0.267)	1.339*** (0.296)
R&D firm alliance experience	0.015* (0.008)	0.029*** (0.008)
R&D firm knowledge stock	0.021*** (0.006)	0.029*** (0.006)
Constant	-2.739*** (0.696)	-2.377*** (0.656)
Years FE	36.15***	51.59***
Therapeutic areas	YES	YES
alpha	3.71	3.04
χ^2	9723.34	12468.60

N= 960. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Supplemental analysis

We seek to investigate the robustness of our results in several ways. First, we test all three hypotheses using alternative frameworks. We check if zero-inflated negative binomial regressions are more appropriate for all three dependent variables, and conduct vuong tests for all three models, which show that negative binomial models are better fitted for our sample. As our second dependent variable, *search breadth*, and our third dependent variable, *search domain*, are highly correlated, we use a bivariate negative binomial model to test for our results in hypothesis 2 and hypothesis 3. We find consistent results as shown in Tables 3 & 4.

Second, we examine the sensitivity of our models to any specific variables such as knowledge stock, alliance experience, as well as technology value variables. We windsorize the variables and the standardized values give qualitatively similar results as those shown in the main tables. We also calculated alternative measures using 3-year, 5-year, and 7-year windows, and results are also consistent.

Discussion

Drawing upon literatures on external knowledge acquisitions and the internal structure of firms, we explore the relationship between the R&D centralization of the client firm and the client firm's knowledge search strategy relative to its alliance partners. The centralized R&D structure, as represented by the presence of an R&D head in the executive team, can help the firm with a smooth communication channel across business units and provide better coordination through a clear reporting hierarchy. Empirical evidence on a large sample of 960 non-equity R&D alliances in the biopharmaceutical industry provides support for our hypotheses. We show that

firms are more likely to search for technical know-how from their alliance partners if the firm has a centralized R&D structure. We further find that centralized firms conduct knowledge search more broadly from their alliance partners, and they are more likely to use such externally sourced technologies in a wider range of therapeutic areas.

Contributions and implications for future research

Our study makes several contributions to research on knowledge exploration and boundary-spanning, as well as research on the internal structure of the firm. First, we bridge these streams of work and offer insights on how centralization of R&D within the client firm (Argyres and Silverman, 2004; Arora et al., 2014) can exert impact on the design of its knowledge search process in strategic alliances (Oxley and Sampson, 2004; Reuer and Devarakonda, 2016; Ryu et al., 2018).

Second, our paper complements existing studies aimed at analyzing the link between R&D centralization and external knowledge acquisition. Firms typically achieve external knowledge acquisition through M&As and strategic alliances (Rosenkopf and Almeida, 2003), and it is often assumed that firms would easily combine such acquired knowledge with its existing knowledge stock. However, as we consider the organization's internal knowledge repository and that knowledge is often segregated across different research units and teams ((Daneels, 2011)), this study extends our understanding of knowledge acquisition and recombination within the organizational structure.

Third, we offer insights on how centralization of R&D within a firm can exert impact on the client firm's knowledge search behavior towards its strategic alliance

partners. By considering how pharmaceutical firms manage R&D and capabilities internally across research units, we find that R&D centralization plays an important role in shaping the client firm's external knowledge search trajectory towards its alliance partners (Arora and Gambardella, 1994; Zheng et al., 2007). Drawing upon both alliance and patent data, our results show that firms with centralized R&D are focused on higher level innovation and thus are more likely to diffuse and recombine external alliance partner knowledge more effectively within their own research units.

In addition, as empirical studies on organizational structures are rare, we provide empirical evidence using data in bio-pharmaceutical industry to proxy for the R&D structure of the firm. We measure R&D centralization based on whether the position of an R&D head is present in the top management team (Lerner and Wulf, 2007). Our findings are also in line with prior research on R&D centralization using alternative measures (Argyres and Silverman, 2004; Arora et al., 2014), demonstrating that firms with centralized R&D are associated with better communication efficiency when it comes to R&D coordination and are more likely to generate innovations with a wider breadth of technological classes and therapeutic areas.

Limitations

In addition to the research possibilities already noted, extensions might also pursue additional research opportunities presented by some limitations of this study. First, we test our hypotheses using patent data in the biotechnology industry, which can only track the exploration patterns of innovations that are successful enough to have resulted in patents. As firms certainly undertake knowledge search that do not

results in granted patents, it would therefore be valuable in future research to measure knowledge search in more detailed and dynamic ways.

Second, due to the nature of our centralization measure, and similarly to other empirical studies that explore the internal organization of firms, our study cannot identify and infer causal relationships between R&D centralization and the firm's knowledge search behaviors. One reason is that the internal management of R&D does not vary much over time, so it is difficult to observe the within treatment effect where one firm switches its internal structure of R&D over time. To alleviate endogeneity concerns, we have created matched samples to examine the effect of R&D centralization on the alliance partner knowledge search, and we have also created alternative measure to explore the knowledge search behavior from multiple angles (search volume, breadth, and domain). Future research can make further attempts at finding alternative centralization measures that change within the firm over time or identifying instrumental variables to derive casual inference of our study.

Additionally, another limitation in our study is we focus on the bio-pharmaceutical industry in our analysis. In this industry, firms involve in complex agreements which might have unique alliance contracting conditions. An extension of our analysis in other hi-tech industries would strengthen our findings. We also restricted our analysis on contracts that are signed between two parties. Future studies could investigate multilateral agreements and extend the analysis.

We hope that our study would encourage more research on exploring the role of R&D centralization on the firm's knowledge search and diffusion process. Our

measure for R&D centralization is based on presence of an R&D head in the executive committee (Lerner and Wulf, 2007). Future studies could look into R&D management alternatives and strengthen the results. For example, prior scholars have examined the role of alliance portfolio managers in governing alliance activities, and future studies can extend our analysis of R&D centralization and test the role and interaction between the R&D head and the alliance portfolio manager. Similarly, scholars have shown that having a steering committee in a non-equity alliance can help mitigate the knowledge misappropriation hazards (Reuer and Deverakonda, 2016). It would be interesting to check the relationship between R&D centralization and the presence of steering committees in R&D alliances in future research.

Conclusion

We bridge current streams of corporate and cooperative strategy research to explore the interplay between the organizational structure of R&D and external knowledge search. We propose that in an alliance between a client firm and an R&D firm, the client firm's internal structure of R&D plays an important role in the knowledge diffusion process of the alliance, specifically in determining the technological sourcing strategy in relation to their alliance partners. Using non-equity alliances data from the bio-pharmaceutical industry, we find that firms with centralized R&D structure tend to search more extensively from alliance partners. We further find that centralized firms conduct knowledge search more broadly from their alliance partners, and they are more likely to use such externally sourced technologies in a wider range of therapeutic areas. We hope that this study encourages more research on organizational structure in the context of R&D

centralization as well as work joining the literatures on external knowledge acquisition and organizational boundary-spanning

CHAPTER 4.

A Signaling Perspective on Employee Mobility

The Darkside of Signaling

Introduction

It has been widely acknowledged that new ventures face major challenges in procuring critical resources necessary to exploit their growth opportunities. In particular, the liability of newness can seriously hamper their ability to access inputs in factor markets as well as meet demands for outputs in product markets, thereby making it difficult to achieve competitive advantage (Stinchcombe, 1965; Delacroix and Carroll, 1983). For instance, the lack of information on raw material, partner, labor, or output market conditions can lead new ventures to make costly mistakes (Gulati, 1999; Nicholson et al., 2005; Shipilov et al., 2005). Moreover, information asymmetry about a venture's technological capabilities makes it challenging for them to obtain financial resources from external capital providers on attractive terms (Amit et al., 1990; Hsu, 2004). In particular, new ventures struggle to attract and retain high-quality employees to enable the organization to develop (Acs, Morck, Shaver and Yeung, 1997). Consequently, new ventures' resource acquisition challenges can have an adverse impact on their ability to survive and grow in a competitive environment with large incumbent rivals (Evan and Jovanovic, 1989; Gimeno, Folta, Cooper, and Woo, 1997). These challenges are particularly significant in high-tech industries given the lack of the new venture's industry track record and the importance of intellectual property and human capital to a technology venture (Stuart, 1998; Stuart, Hoang, and Hybels, 1999; Gulati and Higgins, 2003).

Strategy and entrepreneurship scholars have highlighted the role of signals in addressing these challenges. In the presence of information asymmetry, new

ventures can engage in activities to signal their quality and thereby facilitate economic exchanges in various factor markets (Riley, 2001; Connelly et al., 2011). While there are many potential signals a firm might employ, research on high-tech ventures has emphasized two types of interorganizational relationships as signals for quality and the new venture's future prospects: venture capitalist (VC) prominence (Lee, 2001; Sanders and Boivie, 2004; Cohen and Dean, 2005) and alliance network prominence (Gulati and Gargiulo, 1999; Podolny, 2001). Specifically, the new venture's affiliation with prominent VCs and its network of alliance partners are both effective signals which are costly for new ventures to obtain, yet they convey that the new venture's unobservable resources and prospects are attractive. Given such signaling activities, a new venture can overcome survival problems due to its tenuous resource position and can obtain benefits such as partnerships to access needed resources in various factor markets (Hsu, 2006; Ozmel, Reuer, and Gulati, 2013). Securing access to such partnerships allows the firm to obtain additional resources such as human capital or funding. Earlier research on information economics and signaling theory emphasized that new ventures can access superior physical resources and overcome issues such as financing problems and access to partners (Gulati and Gargiulo, 1999; Stuart et al., 1999; Hsu, 2004). However, another important resource for new ventures is human capital, and it is not quite clear in previous literature what signals might affect the factor market for human capital.

While traditional signaling studies have focused on the effect of signals at the individual level such as how signaling helps individuals to get jobs (Spence 1973), the

effect of firm level signaling on new ventures' human capital retention has generally been underexplored. For instance, Spence's (1973) seminal article on signaling in labor markets shows that employees can use their educational achievement as a signal of their productivity to appeal for positions and higher wages. However, the question of whether and how firm level signals such as VC prominence and alliance network prominence can affect new ventures in the factor market of human capital remains unanswered.

In this paper, we study the counterpart of Spence (1973) in the employment relationship and argue that new ventures' interorganizational relationships can serve as firm level signals in the market for human capital. New ventures that possess such interorganizational relationships would signal that they have secured critical resources such as its research scientists and patent inventors, which potentially boost the visibility and attractiveness of these key human capital. Distinct from other physical or financial factor markets, human capital as a type of resource is indivisible, and the allocation and distribution of such resource across firms are not entirely controlled by the firm (Coff, 1997; Campbell et al., 2012). Inventors and other employees might be reluctant to stay in startups because of the uncertainty about a new venture's prospects relative to their alternative job opportunities. We investigate this particular information asymmetry problem and consider whether signals conveyed by new ventures may in fact make individuals more prone to leaving and start employments elsewhere. We investigate the effect of two signals—VC prominence and alliance network prominence—on employee retention

in the market for knowledge workers. Because of the imperfect information and information asymmetry, new ventures may have a hard time keeping high quality employees. Such employees may prefer employment and contracts with established firms because there are less chances that established firms will go out of business during the employees' tenure.

We develop the idea that new ventures can rely on the prominence of their VC affiliates and alliance network to signal their quality, yet with the presence of strong signals, current employees who work at these new ventures may gain more visibility in the labor market, thus become more likely to be noticed and poached by other employers in the market. Thus, we examine whether such signals are also associated with more employees departing from their focal ventures. This points to a potential downside of signaling that new ventures employing strong signals may risk losing existing human capital at the same time as they attract new talents. We draw on employee mobility literature that shows individual employees carry a significant portion of organizational knowledge with them when they leave, resulting in undesirable knowledge spillover for the new venture. Therefore, it is important to examine separately the effects of the abovementioned signals on both inbound and outbound mobility activities of the new venture.

We conduct empirical analyses in the context of inventor movements in the biopharmaceutical industry. This setting offers several advantages: First, inventors are specialized human capital which is of great importance to the firm. Second, while

change of employment is common in the industry, the use of patent data enable use to track inventor movements relatively accurately. Third, there are high frequencies of VC investments and alliance formations which enable us to observe the signaling events using network measures. Our empirical findings on negative binomial regressions largely support our hypotheses and demonstrate that both signals of VC prominence and alliance network prominence are associated with existing inventors subsequently leaving the focal venture. In addition, our data reveals that these signals are more pronounced when there is a high level of information asymmetry as we find that the signaling effects are reduced with the number of investment rounds.

This study makes several contributions to the literature on signaling and strategic human capital retention for new ventures. First, this paper contributes to the entrepreneurship literature by showing how new ventures may face challenges in keeping important human capital with signaling, and we contribute empirical evidence by showing that new ventures' network affiliations negatively affect the likelihood that new employees are going to stay in the newly established firms. Second, we build an underexplored link between firm level signaling and human capital retention. While Spence (1973) investigates job candidates signaling to hiring firms, our study examines the counterpart of the employment relationship where new ventures engage in signaling through interorganizational ties to attract potential employees in the labor market. Second, while prior studies on signaling theory have identified many benefits of signals, our study points to a potential downside of signaling which has been largely overlooked. Third, signaling theory research has

also suggested that the benefits of interorganizational relationships are likely to depend on the informational environment of signals in different stages of the life of the firm (e.g. Gulati and Higgins, 2003). Our study examines the signaling effects over time and finds that these effects become less pronounced as the new venture goes through multiple investment rounds.

Theory and Hypotheses

Background Theory

New ventures face major challenges in procuring vital financial and other resources to develop their products, and new venture must learn to cope with information asymmetry and asymmetric information in various factor markets in order to profit from their technological resources. First, new ventures lack proven operating history (Stuart et al., 1999), and their core asset bases consist mainly of intangibles such as intellectual property and R&D capabilities. While these intangibles are important indicators for the potential value of the business, their quality and future prospects are difficult to verify until much later stages (Shane & Stuart, 2002; Shane and Cable, 2002). Then, due to appropriation concerns, a new venture can be unwilling to disclose private information which is critical for the venture's valuation and growth (Arrow, 1962; Anton and Yao, 1995). Consequently, new ventures are incentivized to misrepresent information about their resources, products, and operation in order to profit from economic exchanges in the financial and other input markets (Amit et al., 1990). This creates the classical "lemon

problem”, as the new venture’s exchange partners are unable to differentiate ‘good’ and ‘bad’ ventures (Akerlof, 1970).

To lessen these resource constraints, a new venture can engage in costly activities to signal its resources and potential prospects. Spence (1973) originally employs signaling perspective to analyze job applicants in the labor market. He shows that, at the individual level, more productive job applicants need to engage in costly activities such as higher education achievements in order to gain higher salary compared to less productive job applicants. While productivity is not directly observed, employers may infer quality and capability through the signal of a job applicant’s education attainment, because less productive job applicants would find it costly to achieve a high level of education achievement.

Extending signaling theory to the firm level, new ventures with sufficient resources and great prospects can employ costly signals to differentiate themselves from other ventures. Specifically, new ventures may be willing to seek investment from prominence VCs and sell their shares to them at a discount (Hsu, 2004). Alternatively, new ventures may form alliances with reputable partners with less favorable terms (Nicholson et al., 2005). In either case, the new venture’s affiliation with a prominent inter-organization network can help modify its information environment and mitigate adverse selection risks.

While existing literature on signaling theory has extensively analyzed the impact of VC prominence and alliance network prominence on various factor markets

such as financing, liquidity events, partner selection, and contract negotiations (e.g. Stuart et al., 1999; Gulati and Higgins, 2003; Hsu, 2006; Ozmel et al., 2013), it has devoted far less attention to its impact on the human capital market. This represents an important omission because human capital is a critical and distinct type of resource for high-tech ventures. To start with, strategic human capital refers to an individual's stock of knowledge, skills, and abilities (Coff and Kryscynski, 2011), and given the tacit and complex nature of knowledge, a significant portion of organizational knowledge is embedded in the individual employees. When employees move across organizations, they could potentially bring valuable knowledge to their new employers. Moreover, human capital is a distinct organizational resource due to its indivisibility. While a firm may have full control the allocation of its financial and physical resources, it is quite a tricky business to distribute and fit human capital within the organizational structure over an extended period of time, as each employee represents a unique set of knowledge, information, and skills (Lepak and Snell, 1999). In addition, as resources in other physical factor market are generally compliant with organizational directive, human capital constantly suffers from complications with job satisfaction and turnover (Babin and Boles, 1996).

Inventor Mobility

As we discuss below, we extend signaling theory to the strategic factor market of human capital and explore its effects on employee mobility events. Employee mobility occurs when the new venture loses innovative talents switch to and take on a new job offer, or when they leave to found their own startups. Existing literature

have explored the antecedents of talent turnover and have found various reasons and motives behind employee mobility. On the one hand, recruiting key employees from a firm's competitors can become an important way of acquiring external proprietary knowledge. Thus, the new venture that loses key employees face substantial consequences from knowledge leakage (Song, Almeida, & Kogut, 1999; Rosenkopf & Almeida, 2003; Palomeras & Melero, 2010). On the other hand, employees may leave their current employment and seek out better offers (Zender, 1997; Singh & Agrawal, 2010). Such phenomenon is particularly striking in the high-tech sectors such as biopharmaceuticals where organizational capabilities are heavily embedded in the research scientists and inventors of the firm's innovation products. Thus, employee mobility events can be potentially informative and central in shaping the hiring firm's knowledge search trajectory, yet the employee's previous firm may face challenges of unintended knowledge leakage and spillover.

Despite agreeing on the fact that employee capability and employee benefits play a role in determining individual mobility, it is not clear yet as to how they affect employees' decision specifically. While there are many alternative reasons for why individuals would leave a new venture, our study focuses on the signaling effect at the firm level. Our particular interest in employee mobility lies in investigating the impact of signaling on the venture's ability to retain its human capital at the firm level. Given that signals provide a sense of assurance in the presence of information asymmetry, new ventures are assumed to be able to more easily access resources such as specialized human capital in the labor market, and we expect that employees

associated with such signaling ventures would gain higher visibility and certification, thus more likely to obtain outside options and alternative career offers. In the next subsections, we examine two interorganizational relationships as signals, VC prominence and alliance network prominence, and their effects on the mobility of knowledge workers such as research scientists and patent inventors.

Venture Capital Prominence. Because of imperfect information and information asymmetry, new ventures may find it difficult to keep high quality employees. New ventures can rely on the prominence of their VC affiliates to signal their quality (Gulati & Higgins, 2003), suggesting that they can offer superior resources and benefits to attract human capital. Affiliating with a prominent VC can convey such signals for several reasons. First, VCs generally invest in new venture in the presence of information asymmetry concerning its technological resources and capabilities, so it is critical for VCs to conduct due diligence and collect information regarding these resources and their growth opportunities (Carter & Manaster, 1990). Before investing, VCs form syndicates to share information and their evaluations of the new venture with each other (Li, 2008). Thus, a new venture with prominent VC affiliations conveys quality signals through the extensive information channels and due diligence conducted by the VC syndicates (Hsu, 2006; Koka and Prescott, 2008).

Second, prominent VCs are highly selective and they go through careful screening processes to pick out a small number of investment targets. Because prominent VCs have high visibility and good reputation as successful investors, they

try to maintain their reputation by weeding out low-quality ventures prior to investing (Poldony, 1994; Hsu, 2004). Therefore, new ventures affiliated with them are highly valued, which is also why new ventures are willing to sell their shares to prominent VCs at a discount. The most prominent venture capitalists will also be selective in investing in startups over time, so their ongoing affiliation with a startup can provide assurances that the startup continues to make good progress in its development of its technologies and products. This means that having a prominent VC backing a firm provides a powerful certification of the firm's quality.

Third, new ventures must be able to bear the high cost in order to obtain affiliations with prominent VCs (Hsu, 2004), and high-quality ventures are more likely to afford such costs than lower quality ventures. The new venture is in effect paying for a lease of the reputation of such VCs so as to secure valuable resources from outsiders in the future, whether external financial investors, other partners, or important human capital.

Overall, these arguments above suggest that prominent VC affiliations signal that the venture has high quality resources and superior prospects. The foregoing discussion highlights that there is significant research investigating how startups obtain follow-on benefits from such affiliations in various market contexts (e.g., IPOs or acquisitions involving startups, alliances, etc.), and we extend this argument to the labor market context for knowledge workers. So, it can be inferred that new

ventures associated with such affiliations would employ high quality knowledge workers that are top notch inventors with good technological prospects.

Therefore, we expect that the affiliation with prominent VCs may influence employees' decision to leave the firm via two mechanisms. First, due to careful screening and selection, new ventures associated with prominent VCs are perceived to have better access to resources such as human, intellectual, alliance, and financial capital (Baum et al., 2000). Thus, employees at the new venture enjoy such inferred signaling effect that they are highly valuable and specific human capital. Second, if the new venture is affiliated with prominent VCs, then the new venture gains greater visibility in the market. That is, the new venture's key employees, technology, operations, and business models are more likely to be seen by other ventures in the market and all its information is more likely to be included in financial analyst coverages (Jain & Kini, 1995). Then, other firms in the market would observe the signal about the quality and prospects of the focal venture, and employee poaching from the new venture would be more likely to ensue. Thus, we specify the following hypothesis:

H1. A new venture's affiliation with prominent VCs will have a positive impact on the number of employees leaving the new venture.

Alliance Network Prominence. The prominence of the new venture's alliance network can similarly convey signals on the new venture's resources and prospects (Jensen, 2004) and have a bearing on the startup's sourcing of knowledge

works. In general, alliance partners themselves face information asymmetry when making partnering decisions with new ventures. To alleviate problems associated with adverse selection, alliance partners also carry out due diligence and evaluations prior to alliance formations as well as during their process of engagement with a startup (Nicholson et al., 2005). The fact that alliance partners have vetted the startup, and these partners themselves have alternative options for forming alliances, indicates that the startup's resources and capabilities, including their important human capital, are in demand by other organizations who see value in collaborating with it. Thus, the fact that the new venture is able to form extensive ties in the alliance network serves as indicator for the quality of its resources (Jensen, 2003; Ozmel et al., 2013; Hoang, Hybell, & Stuart, 1999) and human capital. Thus, alliance partners can provide a valuable certification of the new venture that accrues positive evaluations from other knowledgeable actors in the market, similar to the arguments for VCs in the foregoing discussion.

Given that alliance network prominence conveys a new venture's resources and prospects and hence mitigates adverse selection risk, the new venture's prominent position in its alliance network indicates a higher level of visibility in the market (Jensen, 2003; Gulati & Higgins, 2003; Ozmel et al., 2012). Individual inventors from the focal venture would enjoy a more extended network developed by their current employers, which means that they tend to enjoy a wider range of career visibility and outside options. Moreover, by engaging in joint activities with extensive alliance partners, individual inventors are likely to be perceived as well-connected and

adaptive to varying working environments, thereby conveying a high-quality signal. Thus, we expect that a new venture's alliance network prominence is positively linked to the new venture's employee mobility as well.

H2. A new venture's prominence in its alliance network will have a positive impact on the number of employees leaving the new venture.

The Contingent Effects of Signals

The role of the two interfirm relationships discussed—VC backing and alliances—can help resource providers to correctly infer the quality of a new venture. As the level of information asymmetry between the new venture and others increases, the strength of the relationship between the signals and the hiring of new employees strengthens. However, when the information asymmetry is reduced, the impact of the signals weakens. As a firm shows a longer track record, information asymmetry about the firm's prospects decreases, and the effects of the early signals diminish (Certo, 2003; Heeley et al., 2007). For instance, Stuart et al. (1999) find that the relative weight given to a startup's early signals as opposed to a startup's actual performance, should decrease with the age of the firm. This phenomenon can also be observed in the terms that new ventures receive from investors as the new ventures' owners accumulate experience. Kaplan and Stromberg (2004) show that new ventures receive better terms in financing rounds as the owners' experience increases.

We argue that the level of information asymmetry between the new venture and the labor market determines the strength of the previously hypothesized effects. Specifically, we expect that as the new venture's track record increases, signals would

play a lesser role in reducing the risk of adverse selection from the perspective of the rival firms that can potentially poach employees from the new venture. There are two main reasons to explain why new venture will experience a weaker signaling effect as the information asymmetry is reduced. First, the new venture's track record would demonstrate the extent to which the venture is going to grow in the following years. When new ventures have demonstrated that they are able to secure multiple financing rounds, rival firms and other players in the market for human capital can infer the quality of new venture employees by their actual work outputs, such as patent grants and product development, rather than the inferred quality through signaling. Second, evidence shows that new ventures tend to grow considerably after the first few years of operations (Acs et al., 1997). Due to the new venture's growth and their technological progress materializes over multiple rounds, employees in later rounds do not enjoy the same level of inferred visibility or the same amount of outside options as they would in earlier investment rounds.

In this study, we use the number of current investment rounds as a proxy for the new venture's track record. Because investment entities stage their capital infusions and they carefully monitor and reevaluate the new venture in each stage, ongoing backing by a prominent VC conveys actual information on the venture instead of signals. As a new venture goes through multiple VC investment stages into later rounds of investments, the new venture gives out more information about the firm, thus reducing information asymmetry. Therefore, applying this line of

argument to the signaling roles of the above inter-organizational relationships examined earlier, we specify the following predictions accordingly:

H3. The positive effects of a new venture's VC prominence on its employee mobility will be reduced as the new venture's track record increases.

H4. The positive effects of a new venture's alliance network prominence on its employee mobility will be reduced as the new venture's track record increases.

Methods

Sample and Data

We collected the data from multiple sources. First, we gathered a sample of venture capital-backed startups that received their first venture funding between 1980 and 2004. This is because VentureXpert's data is less complete prior to 1980, and we also wanted to have a time window to track the follow-on consequences of signals and investigate the development of new ventures. For this research, data are drawn from Thomson Reuters' VentureXpert database (Katila, Rosenberger, and Eisenhardt, 2008; Lerner, 1995; Sorenson and Stuart, 2001). We also used VentureXpert's classification for industry subgroups and their assignment of ventures into seven subfields: Biosensors, Biotech Equipment, Biotech Research, Biotech Animal, Biotech Human, Biotech Industrial, and Biotech Other.

We then built the careers of inventors employed by U.S. biopharma firms. In order to identify the inventors and the firms, we obtained a list of patent assignees that had patents classified in any of the biopharma technological classes. Then, we identified the U.S.-based firms using the country field of the patent applications from

different data sources. We also cross checked the list of US-based assignees with names of firms based on the U.S. from several sources. Using the SIC codes of the biopharma industry (i.e. 2833, 2834, 2835, 2836, 8731) we retrieved all the firm names from Compustat, SDC M&A data, and VentureXpert. We also collected the names of firms involved in alliances from Recap. After merging the data from all these sources, we compiled a list of firms from the U.S. that compete in the biopharma industry.

Next, we obtained the inventor data from the inventor name disambiguation project by Li et al. (2014). Using Li and colleagues' data, we were able to identify all the patents of each inventor. Even though we did not have to disambiguate the names of inventors, we had to disambiguate assignee names. Since we were interested in finding out the inventor's employer for each year, we had to correctly record the different variations of an assignee name. Combining manual checks with computational algorithms, we created a dictionary of all the assignee names in our sample. We created this dictionary to group the assignee names that identified the same firm but that were spelled in different ways. Using this dictionary, we created a career table in which each inventor was associated to an assignee each year.

Finally, we matched the VC-backed ventures to the assignee names from the inventor career data. We then aggregated all individual inventor-year level data up to the firm-year level. In the context of new ventures, initial public offers (IPOs) and mergers and acquisitions (M&As) are strong signals of the firm's quality or result in

the firm no longer operating as an independent organization. In order to isolate the effect of VC prominence and alliance formation, we run the analyses using the years previous to a new venture's IPO or to the acquisition of the new venture. Therefore, our final sample consists of a panel dataset of 1709 firm-year observations, and we engage in firm fixed effects negative binomial models to test our hypotheses.

Variables

Dependent variable. The dependent variable in our analysis is *mobility*, and we measure it as the count of inventors who have moved out of the focal firm in a given year. That is, *mobility* is the number inventors who have left the firm in that year. In order to compute the dependent variable, we first need to record whether each individual inventor has moved in the given year. Because identifying mobility events is a key part of this study, we took several steps to make sure that mobility events were correctly identified. The main issue is that while a mobility event involves the change of the employer's name from one year to the other, not all the changes of employer's name indicate a mobility event. In other words, the change of name is a necessary but not sufficient condition for establishing mobility between two firms. As an illustration, in 2002 American Home Products Corporation (AHPC) changed its name to Wyeth. Thus, inventors employed by AHPC might appear as they moved if they are later employed by Wyeth, but they stayed at the same firm. Therefore, we searched the web in order to obtain information regarding a change of employer's name. We concluded that a change of name was a mobility event when

we did not find any evidence of a corporate event that would be conducive to a firm changing its name.

Independent variables. We measured *Venture Capitalist Prominence* via the eigenvector centrality of the VCs in the VC syndicate networks (e.g., Bonacich, 1987; Podolny, 1993 & 1994; Ozmel et al., 2013). The centrality of VC firm i in year t is measured using all of the direct and indirect syndicate ties formed between VC firm i and all the other VC firms during the most recently past five years (between $t - 5$ and t), as shown below:

$$VC\ Prominence_{i,t} = \sum_{j=1}^{N_t} (\alpha_t + \delta_t c_{j,t}) R_{i,j,t} \quad (1)$$

where $c_{j,t}$ is the centrality of the VC firm j in year t ; $R_{i,j,t}$ is the relationship matrix entry indicating the number of co-investments between VC firm i and VC firm j during the last five years; δ_t is the weighting coefficient, which is set equal to three quarters of the reciprocal of the largest eigenvalue of R (e.g., Podolny, 1993; Jensen, 2003); and α is the scaling factor, which assures that the maximum centrality for each year is equal to 1 across all the VCs. If there are multiple VCs investing in a particular technology venture, we picked the maximum of those VCs' centralities (Ozmel and Guler, 2015), but we also investigated the mean value to assess the sensitivity of the findings and results are qualitatively similar.

Our second theoretical variable is *Alliance network prominence*. We measured the focal venture's alliance network prominence based on its eigenvector centrality in the alliance network (Baum and Silverman, 2004; Ozmel et al., 2016; Bonacich, 1987),

using data from Thompson Reuters Recap on alliances formed in the biotechnology industry:

$$Alliance\ Network\ Prominence_{i,t} = \sum_{j=1}^{N_t} (\alpha_t + \delta_t c_{j,t}) R_{i,j,t} \quad (2)$$

where $c_{j,t}$ is the centrality of the new venture i 's alliance partner, established company j , in year t , and $R_{i,j,t}$ is the relationship matrix that shows the number of alliances formed between i and j during the last five years, up to year t . We normalized the centrality measure so that the maximum centrality in a particular year is equal to 1.

In addition, we measure the new venture's track record as *the number of investment rounds* in which the new venture had been involved (Stuart et al., 1999). Because VCs stage their capital infusions, and they carefully monitor and reevaluate the new venture in each stage, the development and progress of a new venture is reflected in the number of rounds of investment it has received. At each round, additional information is generated on its technologies and products, thereby reducing information asymmetries between the new venture and prospective exchange partners.

Control variables. In order to address the potential of alternative explanations to the number of newly hired employees, we obtain characteristics of the new venture's CEO and top management team, the firm and the firm's assets, and of the VC firms that have invested in the venture.

CEO and top management team. Employees may focus on the experience of the managers in charge of the new venture when deciding to stay in the new venture. The managers' characteristics are likely to determine the potential to secure funding in the future, which is critical for the survival of a young venture. We control for the *CEO Age* because it can capture the experience of the CEO (e.g., Buchholtz, Ribbens, and Houle, 2003; D'Aveni and Kesner, 1993). We also control for the *CEO Industry Experience* because it captures the extent to which the CEO is an expert in the venture's industry, which can determine the access to resources (e.g., Cohen and Dean, 2005; Shane and Stuart, 2002). Finally, we control for the *Top Management Team (TMT) Size* because larger teams may be better equipped to manage firms in uncertain conditions (e.g., D'Aveni and Kesner, 1993). Data for the above variables were hand-collected from multiple sources, including Mergent Online and Bloomberg Business, and VentureXpert.

Venture's characteristics. We also control for the venture's *Firm Age* because it helps to capture the extent to which a venture faces the liability of newness (Stinchcombe, 1965). As ventures become older, the chances of surviving increase. We also include measures of the venture's future prospects that can also explain why some employees may decide to join the venture. Specifically, we control for the quality of the venture's knowledge because the venture's future prospects partially depend on the venture's current knowledge. A venture's technological accomplishments through its patenting activity is measured by its *Citation-Weighted Patent Stock* by tracing the number of forward citations that a patent received in the

subsequent years (e.g., Griliches, 1981; Hall, Jaffe, and Trajtenberg, 2005). All the patent information was collected from the USPTO database. We also control for the *Pipeline* of the new venture by measuring the number of products at preclinical, clinical, and commercialization stages respectively. This variable helps us to take into account the likelihood that the venture is going to keep developing valuable products in the future.

VC firms' characteristics. We control for the *VC's Early Investment Stage* because it serves as a proxy for the ability of venture capital firms to manage investments in risky ventures. We measured this variable as a dummy that equals one when the venture is backed by a VC fund focusing on the early stage in VentureXpert's classification. In addition, our models also include years and industry subgroups fixed effects that help us to account for time trends and industry characteristics.

Analysis and Results

Statistical Analyses

We employ a firm-year fixed effects negative binomial framework for our regression of inventor mobility because of the count nature and over dispersion of mobility events. The dispersion parameter is significant in all our models, thus confirming over dispersion in the data and supporting the use of the negative binomial models. We report heteroskedasticity-robust standard errors. Tables 1a and 1b present descriptive statistics and the correlations matrix. We exclude observations where firms go through initial public offerings, acquisitions, or becoming

living dead, and the resulting sample consists of 1709 firm-year observations in our sample with a reasonable amount of mobility events happening across the years. At a given year, the maximum number of employees joining a new venture in our sample is 10, and the minimum is 0 meaning that no scientist is hired. Similarly, there is maximum of 12 employees leaving the firm in a year, and a minimum of 0 leaving the firm. This means that inventors may potentially change employers either individually or in teams. We also examine variance inflation factors (VIF) to diagnose multicollinearity concerns and find that the mean VIF is less than 3 for all models, which is well below the critical value of 10.

Table 4-1a. Variable Definition and Descriptive Statistics

Variable	Definition	Obs	Mean	Std. Dev.	Min	Max
Mobility	Number of inventors leaving the focal firm in a given year	1,709	0.155	0.651	0.000	12.000
Alliance network prominence	Eigenvector centrality of the focal firm in its alliance network	1,709	0.020	0.055	0.000	1.000
VC prominence	Eigenvector centrality of the focal firm's primary VC	1,709	0.185	0.309	0.000	1.000
Number of current rounds	Number of current round of VC investment	1,709	2.641	1.921	1.000	13.000
Earlstage	Early stage of round (0/1)	1,709	0.505	0.500	0.000	1.000
Number of deals managed by VC	Number of deals managed by the company's venture capitalists in the past 5 years (mean)	1,709	60.112	108.892	0.000	825.000
Round amount	Amount of VC funding up to current round	1,709	0.165	0.248	0.000	1.957
CEO age	Age of the company's CEO	1,709	44.977	7.537	25.000	71.000
CEO experience	Industry experience of the company's CEO	1,709	6.377	7.165	0.000	45.000
TMT size	Number of TMT members in the company	1,709	4.990	3.205	1.000	18.000
Firmage	Company age	1,709	9.524	36.598	0.000	422.209
Patent stock	Citation weighted patent stock in the past 5 years	1,709	1.775	8.030	0.000	117.000
Pipeline	Number of products at preclinical, clinical, and commercialization stages	1,709	0.941	2.777	0.000	28.000

Table 4-1b. Correlation Table

	1	2	3	4	5	6	7	8	9	10	11	12
1 Mobility	1											
2 Alliance network prominence	0.2102	1										
3 VC prominence	0.0613	0.1665	1									
4 Number of current round	0.1021	0.1689	0.3591	1								
5 Earliest stage	-0.0187	-0.084	-0.147	-0.4231	1							
6 Number of deals managed by VC	-0.0053	-0.0452	0.0005	0.0761	-0.0588	1						
7 Round amount	0.0587	0.1325	0.2801	0.374	-0.2979	0.164	1					
8 ceo_age	-0.0321	-0.0567	-0.0953	-0.06	-0.0034	0.0108	0.0034	1				
9 ceo_index	-0.0279	0.0191	0.015	-0.0756	-0.0937	0.0504	0.1071	0.5105	1			
10 tmt_size	-0.0235	0.1204	0.1029	-0.0048	0.0051	0.003	0.1775	0.1425	0.252	1		
11 Firmage	0.016	0.007	0.1357	0.1339	-0.0707	-0.0121	0.1203	-0.006	-0.0292	-0.0325	1	
12 Patent stock	0.2599	0.2266	0.0759	0.1454	-0.0708	0.1095	0.2042	-0.0158	0.0367	-0.0782	0.0056	1
13 Pipeline	0.261	0.2387	0.1064	0.1127	-0.0412	0.0224	0.3065	-0.0352	-0.0035	0.0914	0.1426	0.2718
N=1709.												

Table 4-2. Negative Binomial Regressions on Mobility

Variables	(1)	(2)	(3)	(4)	(5)
Alliance network prominence		9.794*** (2.371)	15.573*** (4.298)	9.989*** (2.168)	14.029*** (4.347)
VC prominence		0.698** (0.306)	0.610** (0.310)	1.971*** (0.604)	2.007*** (0.636)
Number of rounds	0.216*** (0.059)	0.145*** (0.051)	0.209*** (0.053)	0.262*** (0.054)	0.268*** (0.060)
Alliance network prominence x Number of rounds			-1.481** (0.721)		-1.258* (0.764)
VC prominence x Number of rounds				-0.412*** (0.146)	-0.375** (0.152)
Early stage VC fund	0.134 (0.274)	0.159 (0.251)	0.236 (0.248)	0.224 (0.261)	0.281 (0.254)
Cumulative VC amount	0.285 (0.378)	-0.161 (0.441)	0.047 (0.424)	0.160 (0.447)	0.415 (0.442)
Pipeline	0.017 (0.083)	-0.018 (0.091)	-0.031 (0.090)	-0.029 (0.083)	-0.006 (0.082)
Citation weighted patent stock	0.026* (0.014)	0.015 (0.012)	0.012 (0.012)	0.011 (0.010)	0.014 (0.011)
1.clinical_stage	0.159 (0.529)	0.310 (0.536)	0.257 (0.537)	0.434 (0.519)	0.229 (0.508)
2.clinical_stage	-0.731 (0.558)	-0.857 (0.744)	-0.840 (0.699)	-0.521 (0.668)	-0.893 (0.622)
3.clinical_stage	-0.794 (0.491)	-0.425 (0.520)	-0.421 (0.529)	-0.546 (0.480)	-0.592 (0.518)
CEO age	-0.023 (0.020)	-0.016 (0.019)	-0.020 (0.019)	-0.039* (0.021)	-0.023 (0.020)
CEO industry experience	0.002 (0.018)	-0.015 (0.017)	-0.017 (0.017)	-0.000 (0.017)	-0.017 (0.017)
TMT size	-0.015 (0.038)	-0.014 (0.033)	-0.008 (0.033)	-0.014 (0.034)	-0.017 (0.032)
Firm age	0.001 (0.001)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.002 (0.002)
Year fixed effects	49.67***	32.86***	20.42***	39.97***	35.52***
Industry Subgroups fixed effects	13.83**	12.19*	12.60**	16.29**	16.48**
Constant	16.675*** (0.613)	25.648*** (6.962)	-30.639*** (10.911)	25.862*** (7.780)	29.181*** (8.767)
Log likelihood	-532.226	-522.485	-520.332	-518.916	-518.073
χ^2	103.03	122.6	124.96	127.79	129.48
α (dispersion parameter)	4.5	3.663	3.617	3.56	3.528
N=1709. *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses					

We now turn to regression analysis to examine the effect of signaling on the new venture's human capital attainment. Table 2 shows estimates of negative binomial regression models where the dependent variable is *mobility*, which is the count of employees leaving the focal venture. As Table 2 shows, column (1) is the baseline model with just control variables, column (2) shows the main effects without interactions, and columns (3)- (4) show interactions between each signal variables and the number of round as a proxy for the level of information asymmetry. Finally, column (5) includes the full set of interactions.

We begin with the results of the main effects model shown in the column (2) of Table 2. In these main effects models, both Hypotheses 1 and 2 are supported, as both alliance network prominence and VC prominence are significant with $p < 0.01$. In particular, the effect size for alliance network prominence is much greater than that of VC prominence. Moving to column (2) of Table 3, we find that the main effects models on column (2) are significant for *mobility*. Thus, Hypotheses 3 and 4 are also supported.

We next investigate the effect of alliance network prominence and VC prominence when the level of information asymmetry of the new venture diminishes. Columns (3)-(4) in both tables show that there is indeed a weakening effect when either or both signal variables are met with reduced information asymmetry. Columns (5) in both tables show that the signaling effects are still quite outstanding and significant even in the later rounds of investment stages. The results and signs

of the coefficients of our key independent variables are consistent across all models, pointing to a downside of signaling that inventors from ventures associated with such signals are more likely to leave the focal ventures.

To further investigate the interaction effects in nonlinear models such as ours (Greene, 2011; Zelner, 2009; Hoetker, 2007), we create two simulation-based interaction figures and illustrate the results of the contingent effects of both signal variables in Figures 1 and 2. We observe downward slopes in both figures, consistent with the diminishing signaling effects of VC prominence and alliance network prominence on mobility as the number of investment rounds increases. Specifically, Figure 1 indicates that the effect of VC prominence is positive yet diminishing till the seventh investment round, which is in line with the expectation that the VC-related effects tend to be more pronounced in the earlier stages of venture growth. Similarly, as shown in Figure 2, the effect of alliance network prominence is quite pronounced in the early stages till the sixth round, and then start to diminish. This may be interpreted as a longer lasting signaling effect through the new venture's own network of alliance partners compared to that of its venture capitalists.

Figure 4-1. Interaction Effects of VC Prominence and Number of Rounds on Mobility

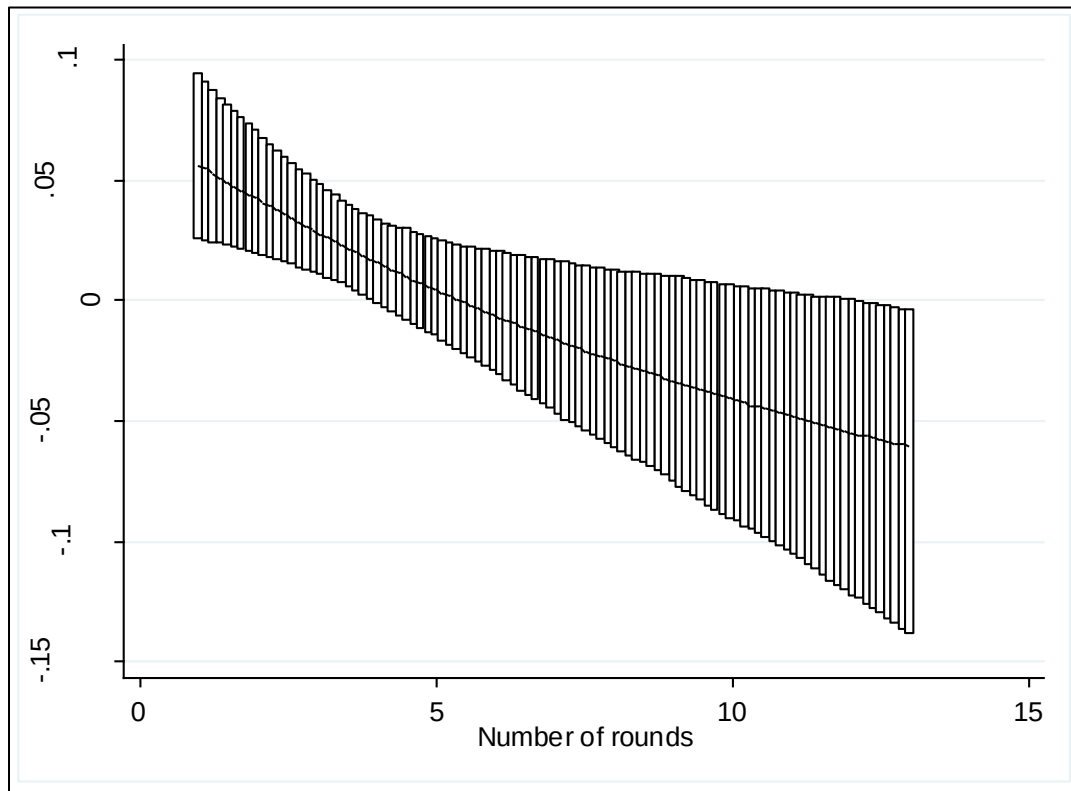
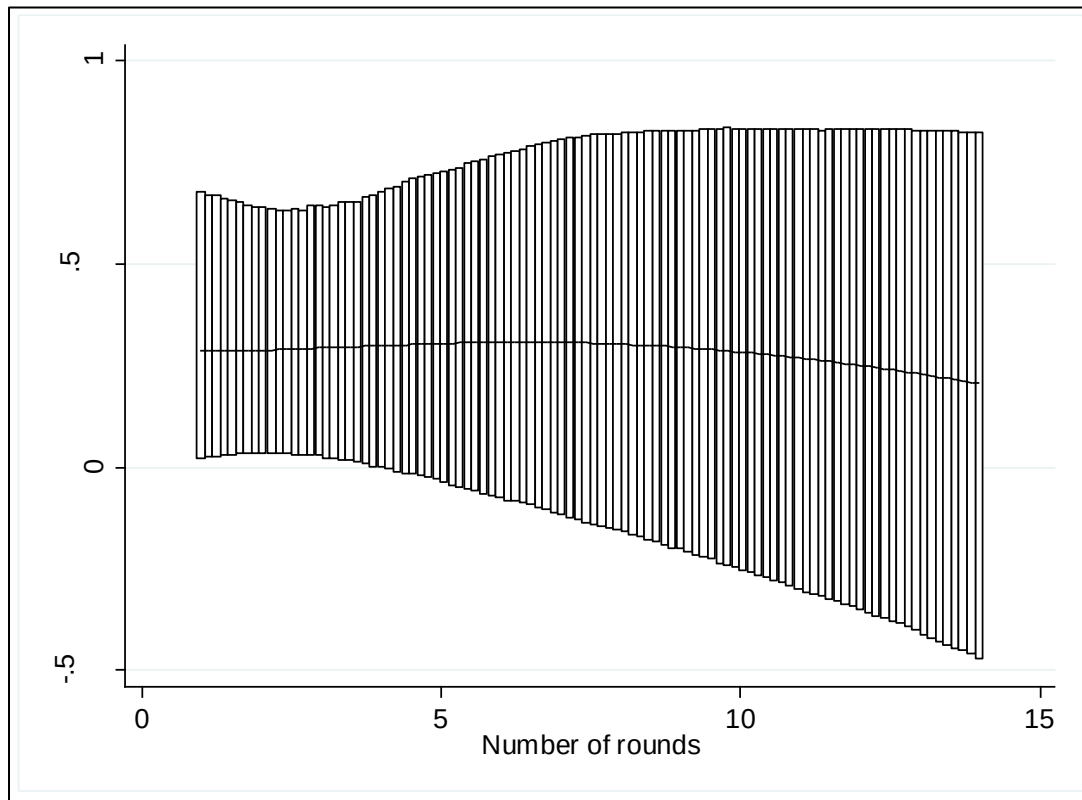


Figure 4-2. Interaction Effects of Alliance Network Prominence and Number of Rounds on Mobility



Supplementary Analyses

We also seek to investigate the robustness of our results in several ways. First, we assess whether the findings and interpretations are potentially subject to the statistical methods chosen above. In addition to the negative binomial framework, we also run our analysis with a logit framework investigating whether or not there is mobility occurring in the given year. The results of these analyses are consistent with the results shown above in Table 2.

Second, we examine if potential unobserved heterogeneity surrounding the interorganizational ties might affect the results and interpretations. We examined the endogeneity of the VC prominence variable by using a two-stage model controlling for self-selection. The first-stage model assesses the VC-backing using state dummies for instrumental purposes, and the resulting second-stage model shows that the self-selection term is insignificant (Baker and Gompers, 2003).

Third, we examine if the effects of alliance network prominence on inventor mobility are due to the signaling mechanism rather than other potential explanations such as inventors poached directly by alliance partners. Thus, we excluded all mobility events wherein an employee leaves the new venture and directly join the alliance partner firm, and we obtained the qualitatively similar results and interpretations as those presented above.

Discussion

Contributions and Implications

By exploring the effect of signals on inventor mobility of new ventures, our study makes several contributions to the literature. First, we build an underexplored

link between firm signaling and human capital retention. Because small and medium sized ventures play an important role in the process of creative destruction, it is important for understanding how ventures can overcome barriers to efficiently attain human capital from external labor markets and thereby prevent unintended knowledge spillover (Campbell et al., 2012; Gambardella and Giarratana, 2010; Acs et al., 1997). There have been few studies that look at implications of inventor mobility in the context of new ventures connected with VCs and strategic alliance partners, and our study explores the market push and pull that drive inventor mobility from a signaling perspective. By focusing on new venture's interorganizational relationships and their effects in the labor market, we respond to calls for research on the antecedent of employee mobility and strategic human capital acquisition (Tzabbar, 2009; Groysberg et al., 2008; Coff, 1997).

Second, our study also advances research on information economics and signaling theory by identifying signals that new ventures send out to the labor market. Previous economics literature on information economics has elucidated the effects of signaling on job seekers at the individual job seeker's level (Spence, 1973), and our study looks at the opposite where new ventures can engage in firm-level signaling to positively affect employee departure.

Furthermore, our findings point to a potential downside of signaling that new ventures may risk losing existing human capital and consequently cause undesirable knowledge spillover, while conventional research often regard signals as valuable and

beneficial assets to new ventures (e.g. Lee, 2001; Gulati and Gargiulo, 1999; Stuart et al., 1999). Management research tends to focus on how firms use interfirm ties as signals to access resources in financial markets as well as partner selection (e.g. Riley, 2001; Connelly et al., 2011), but rarely have scholars discovered potential risks and drawbacks of signaling. We hope that this study encourages more research on signaling theory, particularly in the confluence of literatures on inter-organizational relationships (such as VCs and alliances) and its combined benefits and risks towards the human capital acquisition and retention in the future.

Finally, research on signaling theory has also suggested that the benefits of interorganizational relationships are likely to depend on the informational environment of signals in different stages of the life of the firm (e.g. Gulati and Higgins, 2003). The contingency perspective we develop suggests that the signaling value of interorganizational ties in the labor market varies across different stage of the new venture's funding rounds. We find that the effects of such signals are greater when new ventures are at the earlier financing stages, and the signal value of the new venture's alliance network prominence is longer lasting compared to that of VC prominence in later rounds. We interpret this set of findings as the decaying of signaling effects as information accrues over time. As the new venture enters further into multiple investment rounds, the gap of information asymmetry reduces between the new venture and the labor market, and we have shown that the effect of signaling in human capital attraction changes in different stages of life of new ventures.

Limitations and Future Research Directions

Our study has several limitations that may also present fruitful avenues for future research. To begin with, our study has focused on the attainment of human capital on the firm level. However, due to lack of personal information such as wage and performance evaluation, our datasets limit us from interpreting the motivation or exact cause behind the knowledge worker's career change. While the mobility research spans a variety of different contexts, our understanding of factors that make key knowledge workers more prone to move and join a different firm is far from complete. While our study looks at how signals help resolve recruitment challenges of new ventures in the high-tech industry, we encourage future studies to investigate scientist retention strategies, and it would be interesting to investigate individual factors that may accrue to their change of employment, e.g. seasoned vs. rookie scientists, prolific vs. non-prolific scientists, and individual vs. team mobility (e.g. Agarwal et al., 2016; Kehoe and Tzabbar, 2014 & 2015).

Another interesting topic is related to legal constraints that may inhibit certain individuals from moving to rival firms. Prior studies have shown that firms and individual employees can benefit from mobility events (Rosenkopf & Almeida, 2003; Corredoira & Rosenkopf, 2010; Shipilov, Godart, & Clement, 2017). However, scholars have also demonstrated the effect that legal constraints can have on the likelihood that an individual is going to move (e.g., Marx, 2011). For example, non-compete agreements and the reputation for litigation toughness may present a constraint on employees leaving their current firms (Younge et al., 2015; Ganco,

Ziedonis, & Agarwal, 2015). Future research could study the extent to which the signaling effects can overcome such legal constraints of mobility.

Further, once inventors leave the new venture, research opportunities exist to examine how new ventures promote knowledge recombination by potentially forming ties through former employees (Rosenkopf and Almeida, 2003), or design governance mechanisms to protect their existing knowledge stock (Ganco, Ziedonis, and Agarwal, 2015). On the one hand, research on human and social capital has shown that new hires have a hard time integrating their knowledge with the firm's existing knowledge stock due to status asymmetry (Tzabbar, 2009). On the other hand, firms that hire scientists face risks of knowledge leakage and spillover through the new hire's network and connection to their former employers, and thus it is also important to study specific governance mechanisms that protect against that. For example, Ganco et al. (2015) find that employers build a reputation for litigation toughness to retain high-value scientists.

Additionally, given our focus on new venture's interorganizational ties as signals, ample opportunities exist to examine other signals or remedies for the challenges of human capital retention. Recent research have explored star founders and patent litigations as signals of firms' innovative prospects and capacity. Research investigating contingencies may also prove valuable in identifying boundary conditions of signals. Additionally, future research might also examine the generalizability of our findings on the value of signals by studying new venture's

human capital acquisition in other countries or other industries. For example, technological entrepreneurship is increasingly prevalent in the machine learning and artificial intelligence industry, and an increasing number of datasets are also available beyond patent data. Research using other samples to study signals and mobility might also be valuable to draw comparisons with existing findings.

Conclusion

We investigate firm level factors from a signaling perspective which complements existing mobility literature on how and why innovative talents leave their focal employers in the high-tech industry. Our study shows that two interfirm relationships - VC prominence and alliance network prominence - can serve as signals and provide a higher level of comfort and assurance in the presence of information asymmetry between the new venture employees and other players in the market, thereby enabling the new venture employees to have better access to the labor factor market and more prone to take on alternative job offers. This points to a potential downside of signaling in terms of out-bound inventor mobility. Empirical evidence from biopharmaceutical startups show that new ventures associated with VC and alliance signals may risk losing valuable existing human capital through employee mobility, despite benefiting in many other strategic factor markets. In addition, we show that the signaling effects become less pronounced as the venture's level of information asymmetry decreases. We hope that this study encourages more research on signaling theory in the mobility context as well as work joining the literatures on human capital management for entrepreneurial ventures.

CHAPTER 5.

CONCLUSION

In my first essay, we extend the theory of information economics to the literature on knowledge search and advance research by examining the role of the acquisition in facilitating the acquiring firm's post-acquisition search towards the acquired firm's alliance network. Specifically, we examine the information intermediation role of the acquisition in addressing information asymmetries between the acquiring firm and the alliance partners of the acquired firm. Drawing on a quantitative analysis of acquisitions and alliances in the biotechnology sector, we develop hypotheses and test how the acquiring firm may cite patents and search for technology belonging to these inherited alliance partners. In this study, we suggest that the acquiring firm obtains private information from the acquisition target and thus becomes aware of the quality of these inherited alliance partners. Such information enables the acquiring firm to tap into the extend knowledge network of the target firm and better access the technological resources beyond its local reach. Furthermore, these benefits of the acquisition will be greater when the acquiring firm has a high level of information asymmetry and uncertainty about its inherited alliance partner. Conversely, technological relatedness and prior direct or indirect ties between the acquirer and its inherited alliance partner would substitute the information intermediation role of the acquisition target, thus attenuating the acquiring firm's post-acquisition search towards its inherited alliance partners.

In my second essay, drawing upon literatures on external knowledge acquisitions and the internal structure of firms, we explore the relationship between

the R&D centralization of the client firm and the client firm's knowledge search strategy relative to its alliance partners. The centralized R&D structure, as represented by the presence of an R&D head in the executive team, can help the firm with a smooth communication channel across business units and provide better coordination through a clear reporting hierarchy. Empirical evidence on a large sample of 960 non-equity R&D alliances in the biopharmaceutical industry provides support for our hypotheses. We show that firms are more likely to search for technical know-how from their alliance partners if the firm has a centralized R&D structure. We further find that centralized firms conduct knowledge search more broadly from their alliance partners, and they are more likely to use such externally sourced technologies in a wider range of therapeutic areas.

In my third essay, we investigate firm level factors from a signaling perspective which complements existing mobility literature on how and why innovative talents leave their focal employers in the high-tech industry. Our study shows that two interfirm relationships - VC prominence and alliance network prominence - can serve as signals and provide a higher level of comfort and assurance in the presence of information asymmetry between the new venture employees and other players in the market, thereby enabling the new venture employees to have better access to the labor factor market and more prone to take on alternative job offers. This points to a potential downside of signaling in terms of out-bound inventor mobility. Empirical evidence from biopharmaceutical startups show that new ventures associated with VC and alliance signals may risk losing valuable existing human capital through

employee mobility, despite benefiting in many other strategic factor markets. In addition, we show that the signaling effects become less pronounced as the venture's level of information asymmetry decreases. We hope that this study encourages more research on signaling theory in the mobility context as well as work joining the literatures on human capital management for entrepreneurial ventures.

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